

Universal Coset Field Theory

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Abstract

We formulate Universal Coset Field Theory (UCFT) as an axiomatic four-dimensional operator-algebraic quantum field theory whose local observable net is conditionally rigidly determined by the Albert algebra $\mathbb{A}$ and the compact EIII geometry of type $\mathbf{E}_6$. The theory is represented by an $\mathbf{E}_6$ -invariant coset sector with vacuum orbit $\mathbf{E}_6/(\mathbf{Spin}(10) \times \mathbf{U}(1))$. The associated Haag–Kastler net contains local coset, gauge, matter, and spectral subalgebras, while the Euclidean functional integral is used only as a generator of Schwinger functions for this operator-algebraic theory. Spontaneous symmetry breaking produces **32** Goldstone directions forming an irreducible $\mathbf{Spin}(10) \times \mathbf{U}(1)$ multiplet with complexification $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ and a unique positive invariant metric. A spacetime-filling coset representation induces a four-dimensional Riemannian metric by pull-back, and the Einstein–Hilbert term appears as a one-loop effective term in the Sakharov–Adler sense. Within the renormalization-group truncation studied, the positive-coupling flow is effective only below the boundary $\epsilon < 1/6$; a heterotic ultraviolet completion is therefore formulated as a matching condition, not as an autonomous coset fixed point. The zeta sector is a commutative spectral subalgebra of the coset observable algebra: a positive Galois-equivariant realization of the shifted zeta divisor in this subalgebra implies the Riemann hypothesis. We conclude UCFT is an algebraic quantum field theory in which geometry, gravity, and arithmetic spectral constraints are emergent from the coset representation of the local observable algebra.

Keywords: algebraic quantum field theory, Haag–Kastler net, exceptional Jordan algebra, $\mathbf{E}_6$, EIII, coset field theory, induced gravity, emergent spacetime, heterotic ultraviolet completion, grand unification, Riemann zeta function, spectral determinant, swampland

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1 Introduction

Exceptional algebraic structures recur throughout high-energy physics. The exceptional Jordan (Albert) algebra $\mathbb{A}$ of 3×3 Hermitian octonionic matrices has automorphism group the compact F_4 , and its norm-preserving reduced structure group is the noncompact real form $E_{6(-26)}$ of the complex Lie type E_6 , the type that preserves its cubic norm [1–3]. All coset constructions in this paper instead use the compact real form of the same complex type¹, acting on the complexified Albert algebra $J_3(\mathbb{O})_{\mathbb{C}}$. It contains $\text{Spin}(10) \times \text{U}(1)$, under which the fundamental **27** branches as

$$\mathbf{27} = \mathbf{16}_{+1} \oplus \mathbf{10}_{-2} \oplus \mathbf{1}_{+4},$$

the **16** of $\text{Spin}(10)$ carrying one Standard-Model family together with a right-handed neutrino [4]. Formally real Jordan algebras were introduced as algebras of quantum observables; $\mathbb{A}$ is the unique exceptional member of their classification. The organizing question is how much four-dimensional field theory is fixed once this exceptional structure and its compact E_6 symmetry are specified.

The main result of this paper is a conditional rigidity statement in the language of algebraic quantum field theory. The primary object of UCFT is a net $\mathcal{O} \mapsto \mathfrak{A}(\mathcal{O})$ of local observable C^* -algebras on the reconstructed four-dimensional spacetime. Once the Albert structure, the compact EIII coset geometry, and the quantum kinematical postulate are fixed, the coset representation of this net has no independent gravitational input. No graviton field, no spacetime metric, and no gravitational action are postulated in that sector. A spacetime-filling representation of the EIII coset induces its metric by pull-back (Theorem 17), and the Einstein–Hilbert term appears radiatively, at one loop, in the Sakharov–Adler sense (Theorem 25).

The construction rests on two postulates. The first postulate posits a maximally symmetric Albert structure; the second posits the usual quantum-mechanical kinematics of a Hilbert-space representation with unitary dynamics. The local observable net is then represented by an E_6 -invariant symmetry-breaking sector on the compact EIII coset. A polynomial potential on $J_3(\mathbb{O})_{\mathbb{C}}$, built from the invariant Hermitian form and the Freudenthal adjugate, is coercive and is minimized on the orbit of unit-norm rank-one elements; the corresponding orbit of complex rays is

$$E_6 \cdot [X_0] \cong \frac{E_6}{\text{Spin}(10) \times \text{U}(1)},$$

the compact Hermitian symmetric space EIII [5], of real dimension $78 - 46 = 32$. After the gauged $\text{U}(1)_{\psi} \subset H$ absorbs the overall phase mode (Corollary 13), the 32 physical Goldstone directions form a single irreducible multiplet of $H = \text{Spin}(10) \times \text{U}(1)$, whose complexification is $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$; the invariant metric is therefore unique up to scale and positive definite.

¹Throughout, E_6 denotes the compact real form of the complex Lie type E_6 . It acts on the complexified Albert algebra $J_3(\mathbb{O})_{\mathbb{C}}$ (the minimal $\mathbf{27}_{\mathbb{C}}$) and is the group entering the EIII coset construction; it does not act on the real slice $\mathbb{A}$. The noncompact form $E_{6(-26)}$ appears only as the reduced structure group of the real Albert algebra. See Remark 4.

The coset representation also supplies the geometric data. A spacetime-filling configuration of the Goldstone fields immerses a four-manifold into EIII; its first fundamental form is a nondegenerate Riemannian metric, so the four-dimensional geometry is induced by the representation rather than postulated independently. The Euclidean Schwinger functions of the observable net are reconstructed by the Osterwalder–Schrader theorem. The one-loop effective action of the coset, gauge, and chiral **16** sectors induces an Einstein–Hilbert term in the sense of Sakharov [6, 7]; the metric enters through this induced action rather than through a fundamental graviton.

Two additional sectoral rigidity tests are analyzed. First, within the stated functional renormalization-group truncation, the coset-field action is an effective description below the cutoff $\epsilon < 1/6$. The positive-coupling flow does not supply an interacting asymptotic-safety fixed point; therefore an $E_8 \times E_8$ heterotic ultraviolet completion is formulated as a matching criterion at the boundary. The corresponding heterotic matching scale is $M_{\text{GUT}} \simeq 2 \times 10^{16}$ GeV, while plain non-supersymmetric Standard-Model one-loop running would cross near 10^{13} GeV. Second, the E_6 branchings supply a grand-unified matter skeleton and anomaly constraints. We also explain how the Riemann zeta function enters as a conditional spectral rigidity criterion for a commutative spectral subalgebra of the coset observable algebra: if the shifted zeta divisor admits a positive Galois-equivariant realization in that subalgebra, then the Riemann hypothesis follows.

The paper is organized as follows. § 2 states the postulates, formulates the operator-algebraic QFT, recalls the Albert algebra and its E_6 symmetry, and constructs the symmetry-breaking potential and its vacuum manifold. § 3 derives the emergent four-dimensional geometry and the low-energy field content represented in the local net. § 4 defines the coset σ -model representation, computes its Euclidean Schwinger functions, and derives the induced Einstein–Hilbert term and the coset mass spectrum. § 5 presents the renormalization-group flow, its $\epsilon < 1/6$ validity window, and the grand-unified matching scale. § 6 explains the zeta function through the coset correspondence and gives the spectral reformulation of the Riemann hypothesis. § 7 states the conditional heterotic matching criterion, records the $\mathbb{Z}_{12}$ orbit test, and formulates the resulting relative Albert–EIII swampland criterion. § 8 separates the unconditional representation-theoretic results from the conditional ones and makes the logical dependencies among them explicit. § 9 closes with a summary of what has been established. Proofs and technical computations are collected in the appendices.

2 Postulates, the Albert Algebra, and Spontaneous Symmetry Breaking

Here, we state the postulates, formulate the local observable net, recall the Albert algebra and the compact E_6 symmetry used in the coset representation, construct the algebraic symmetry-breaking potential, and count the resulting 32 Goldstone directions.

Postulate 1 (Maximally Symmetric Vacuum) *The vacuum order parameter takes values in the complexified Albert algebra $J_3(\mathbb{O})_{\mathbb{C}}$, and its dynamics are invariant under the full compact*

exceptional symmetry selected by the Albert algebra: the compact real form E_6 acting on $J_3(\mathbb{O})_{\mathbb{C}}$.

Postulate 2 (Universal Hilbert-Space Dynamics) *Physical states $\psi \in \mathcal{H}$ evolve through a one-parameter family $\{U(t)\}_{t \in \mathbb{R}}$ of unitary operators satisfying $U(0) = \text{Id}$, $U(t+s) = U(t)U(s)$, and strong continuity.*

Remark 3 **Postulate 2** provides the kinematical Hilbert space and the one-parameter family of unitary operators. The local observable algebra is specified below. Whenever we later write a functional integral, we treat it as a formal generator of connected Green functions for that algebraic theory; its constructive interpretation is through the Osterwalder–Schrader reconstruction theorem applied to Euclidean correlation functions [8].

Remark 4 (Real-Form Conventions) Three distinct objects recur throughout, and we fix their names once. The real Albert algebra $\mathbb{A} = J_3(\mathbb{O})$ has automorphism group the compact F_4 and norm-preserving reduced structure group the noncompact real form $E_{6(-26)}$. The complexified Albert algebra $J_3(\mathbb{O})_{\mathbb{C}}$ carries the minimal $\mathbf{27}_{\mathbb{C}}$ of the complex group of type E_6 . The compact real form, denoted E_6 throughout, acts on $J_3(\mathbb{O})_{\mathbb{C}}$ and on $\mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}})$, preserving the cubic norm N and the invariant Hermitian form; it does not preserve the real slice $\mathbb{A}$, and we never assert that it does. All coset constructions below use the compact E_6 acting on $J_3(\mathbb{O})_{\mathbb{C}}$; the noncompact $E_{6(-26)}$ plays no further role. We also write $\text{Spin}(10) \times \text{U}(1)$ for the maximal-rank subgroup $\text{Spin}(10) \cdot \text{U}(1) \subset E_6$, globally $(\text{Spin}(10) \times \text{U}(1))/\mathbb{Z}_4$; the spinorial $\mathbf{16}$ is a representation of this group, not of $\text{SO}(10) \times \text{U}(1)$ literally.

No additional postulates are introduced. The compact real form E_6 acts on $J_3(\mathbb{O})_{\mathbb{C}}$ and on its projectivization; the closed orbit of this action is the compact Hermitian symmetric space EIII, with stabilizer

$$H = \text{Spin}(10) \times \text{U}(1), \quad \mathcal{M} = E_6/H.$$

Inside $J_3(\mathbb{O})_{\mathbb{C}}$ this orbit is realized as the locus of rank-one elements of fixed Hermitian norm (**Theorem 8**).

UCFT is formulated as an operator-algebraic Euclidean QFT. Let Reg_4 denote the directed set of bounded regular regions in the reconstructed four-dimensional Riemannian spacetime. To each $\mathcal{O} \in \text{Reg}_4$ we assign a unital C^* -algebra of local observables

$$\mathfrak{A}(\mathcal{O}) \subset B(\mathcal{H}),$$

generated by bounded functional calculi of smeared coset fields, gauge-fixed currents, and matter fields supported in $\mathcal{O}$. In the conditional framework of this paper, the assignment obeys the standard Haag–Kastler structural conditions [9–11]:

1. isotony: $\mathcal{O}_1 \subset \mathcal{O}_2$ implies $\mathfrak{A}(\mathcal{O}_1) \subset \mathfrak{A}(\mathcal{O}_2)$;
2. locality after Osterwalder–Schrader reconstruction: observables in spacelike separated Lorentzian regions commute;

3. covariance under the reconstructed spacetime isometries and under the compact internal symmetry $H = \text{Spin}(10) \times \text{U}(1)$;
4. existence of a cyclic vacuum vector $\Omega \in \mathcal{H}$ whose vector state $\omega(A) = \langle \Omega, A\Omega \rangle$ produces the Euclidean Schwinger functions.

The quasi-local observable algebra of UCFT is the inductive-limit C^* -algebra

$$\mathfrak{A} = \overline{\bigcup_{\mathcal{O} \in \text{Reg}_4} \mathfrak{A}(\mathcal{O})}^{\|\cdot\|}.$$

The one-parameter unitary group in [Postulate 2](#) acts by automorphisms $\alpha_t(A) = U(t)AU(t)^{-1}$ of $\mathfrak{A}$, and its self-adjoint generator is the Hamiltonian of the GNS representation associated with the vacuum state ω . Thus the functional integral used below is not an additional postulate; it is a Euclidean device for computing the correlation functions of this operator-algebraic theory.

For the zeta sector, the relevant object is a commutative spectral subalgebra of the coset observable algebra. If Δ_{UCFT} denotes the positive self-adjoint coset mass operator, the algebra

$$C_0(\text{Spec}(\Delta_{\text{UCFT}})) \subset \mathfrak{A}$$

is obtained by continuous functional calculus. The Riemann-hypothesis criterion in [§ 6](#) is precisely a statement about when the arithmetic zeta divisor is realized inside this positive spectral subalgebra.

Theorem 5 (Jordan Structure and Dimension) *The Albert algebra*

$$\mathbb{A} = \{X \in M_3(\mathbb{O}) : X = X^\dagger\}, \quad X \circ Y = \frac{1}{2}(XY + YX),$$

is a 27-dimensional real Jordan algebra; it is formally real and exceptional [[1](#), [3](#)].

Proof See [Appendix A](#). □

Theorem 6 (Invariant Forms) *The real Albert algebra admits the trace form and the cubic norm*

$$Q(X) = \text{Tr}(X^2), \quad N(X) = \det_{\mathbb{O}}(X).$$

The form Q is preserved by the automorphism group F_4 , while N is preserved by the norm-one reduced structure group, the noncompact real form $E_{6(-26)}$. On the complexification $J_3(\mathbb{O})_{\mathbb{C}}$, the compact real form E_6 preserves the cubic norm N and the Hermitian trace form

$$\langle X, Y \rangle = \text{Tr}(X \circ \bar{Y}),$$

where $\bar{Y}$ is complex conjugation with respect to the real form $\mathbb{A}$. It preserves no complex-bilinear quadratic form, since the **27** is not self-dual.

Theorem 7 (Exceptional type [[2](#)]) *The reduced structure algebra of $\mathbb{A}$ has complexification of type $\mathfrak{e}_6$. In the compact coset construction below, E_6 denotes the simply connected compact real form of this Lie type.*

Theorem 8 (Closed EIII Orbit Criterion) *Let $X^\sharp$ denote the Freudenthal adjugate on $J_3(\mathbb{O})_{\mathbb{C}}$, characterized by $X^\sharp = 0$ exactly on the elements of Jordan rank ≤ 1 [2, 3]. For $v > 0$ the compact E_6 acts transitively on the rank-one locus of fixed Hermitian norm,*

$$S_v = \{X \in J_3(\mathbb{O})_{\mathbb{C}} : X^\sharp = 0, \langle X, X \rangle = v^2\},$$

with vector stabilizer isomorphic to $\text{Spin}(10)$ and line stabilizer $H = \text{Spin}(10) \times \text{U}(1)$. Consequently

$$S_v \cong E_6 / \text{Spin}(10), \quad [S_v] = E_6 \cdot [X_0] \cong E_6 / H = \mathcal{M},$$

the second orbit being the closed orbit EIII $\subset \mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}})$.

Corollary 9 (Existence of an $\text{Spin}(10) \times \text{U}(1)$ Stabilizer) *In the projectivized complex Albert representation $\mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}})$, there exists a point $[X_0]$ whose E_6 -stabilizer is $\text{Spin}(10) \times \text{U}(1)$.*

Proof Let $V = J_3(\mathbb{O})_{\mathbb{C}}$ be the minimal 27-dimensional representation of E_6 . The projectivization $\mathbb{P}(V)$ contains the closed highest-weight orbit of E_6 . By the standard description of the compact Hermitian symmetric space EIII, the stabilizer of a highest-weight line $[X_0] \in \mathbb{P}(V)$ is the maximal-rank subgroup $\text{Spin}(10) \cdot \text{U}(1)$, denoted in the global convention of this paper by $\text{Spin}(10) \times \text{U}(1)$. Hence the orbit is $E_6 / (\text{Spin}(10) \times \text{U}(1))$. $\square$

Fix $v > 0$, set $c_2 = v^2$, and choose a rank-one element $X_0 \in S_v$ (Theorem 8). Define, for $X \in J_3(\mathbb{O})_{\mathbb{C}}$,

$$\begin{aligned} P(X) &= \langle X^\sharp, X^\sharp \rangle + [\langle X, X \rangle - c_2]^2, \\ V(X) &= P(X) + \alpha [\langle X, X \rangle - c_2]^k, \end{aligned} \tag{1}$$

with $\alpha > 0$ and even $k \geq 2$. Both summands are E_6 -invariant: the Hermitian form is invariant by Theorem 6, and the adjugate intertwines the **27** with its dual $\overline{\mathbf{27}}$ equivariantly, on both of which the compact form acts unitarily [2], so $\langle X^\sharp, X^\sharp \rangle$ is invariant.

Lemma 10 *$P(X) = 0$ if and only if $X \in S_v = E_6 \cdot X_0$.*

Proof $P(X) = 0$ forces $X^\sharp = 0$ and $\langle X, X \rangle = c_2$, i.e. $X \in S_v$; transitivity on S_v is Theorem 8. Conversely, P vanishes on S_v by E_6 -invariance of both summands. $\square$

Theorem 11 (Rigidity of the Symmetry-Breaking Orbit) *The polynomial $V(X)$ in (1) is E_6 -invariant, coercive, and attains its absolute minimum precisely on the rank-one orbit*

$$S_v = E_6 \cdot X_0 \cong E_6 / \text{Spin}(10),$$

a $\text{U}(1)$ fibration over the coset $\mathcal{M} = E_6 / (\text{Spin}(10) \times \text{U}(1))$.

Proof Invariance was established above. Since $\langle X, X \rangle = \|X\|^2$ is positive definite on $J_3(\mathbb{O})_{\mathbb{C}}$, the even-degree term $\alpha[\langle X, X \rangle - c_2]^k$ dominates as $\|X\| \rightarrow \infty$, so V is coercive. By [Lemma 10](#), $P \geq 0$ with $P = 0$ exactly on S_v ; the additional term is nonnegative and vanishes on the same level set. Hence $V = 0$ precisely on the orbit and is positive elsewhere. $\square$

Theorem 12 (Dimension Count)

$$\dim E_6 = 78, \quad \dim(\mathrm{Spin}(10) \times \mathrm{U}(1)) = 46, \quad \dim \left(\frac{E_6}{\mathrm{Spin}(10) \times \mathrm{U}(1)} \right) = 32.$$

Proof The dimension of E_6 is tabulated in [\[4\]](#). The classical formula $\dim \mathrm{SO}(2n) = n(2n-1)$ gives $\dim \mathrm{Spin}(10) = 45$; adding the one-dimensional $\mathrm{U}(1)$ yields 46. The difference is 32. Correspondingly $\dim S_v = 78 - 45 = 33$: the rank-one orbit of [Theorem 11](#) is a circle bundle over $\mathcal{M}$. $\square$

Corollary 13 (Vacuum Manifold) *On the minimizing orbit S_v the breaking pattern of the global symmetry is $E_6 \rightarrow \mathrm{Spin}(10)$, with 33 Goldstone directions. The fiber direction is the phase of the line $\mathbb{C} X_0$, on which the $\mathrm{U}(1)_{\psi} \subset H$ acts through a nontrivial character; in the representation used from [§ 4](#) onward, H (and in particular $\mathrm{U}(1)_{\psi}$) is gauged, and this single mode is absorbed by the $\mathrm{U}(1)_{\psi}$ gauge field through the Stückelberg mechanism (cf. [§ 8](#)). The physical vacuum manifold is the orbit of complex rays,*

$$\mathcal{M} = E_6 \cdot [X_0] = \frac{E_6}{\mathrm{Spin}(10) \times \mathrm{U}(1)},$$

carrying the remaining 32 Goldstone bosons.

Remark 14 Throughout this paper, we denote by $\mathcal{M}$ the coset manifold $E_6/(\mathrm{Spin}(10) \times \mathrm{U}(1))$.

3 Emergent Spacetime and Field Content

The vacuum manifold $\mathcal{M}$ is 32-dimensional, but a spacetime-filling field configuration immerses a four-dimensional world manifold into it. In this section, we show how a frame and a nondegenerate Riemannian metric arise as the first fundamental form of that immersion, how physical Lorentzian spacetime is recovered by continuation, and how the transverse directions combine with $\mathrm{Spin}(10) \times \mathrm{U}(1)$ gauge connections and fermionic coherent states to furnish the matter content.

Choose a local coordinate patch

$$\{\phi^\alpha(x)\}_{\alpha=0}^{31}, \quad \phi^\alpha(x) : \mathbb{R}^4 \rightarrow \mathcal{M},$$

where $\{x^\mu\}_{\mu=0}^3$ are smooth parameters on the world manifold. Write $\theta = g^{-1}dg$ for the left-invariant Maurer-Cartan one-form on E_6 and let θ_α^a denote its components along the broken generators $\mathfrak{e}_6 \setminus (\mathfrak{so}(10) \oplus \mathfrak{u}(1))$.

Remark 15 Throughout this section, the symbols x^μ serve solely as four real parameters that label the field configuration $\phi^\alpha(x)$. Their physical interpretation as spacetime coordinates is

justified only after [Theorem 17](#), where the pull-back metric is shown to be non-degenerate, together with the continuation to Lorentzian signature discussed there.

Let $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$ denote the unique positive-definite E_6 -invariant metric on the broken subspace $\mathfrak{m}$ ([Theorem 22](#)), where $K_{\alpha\beta}$ is the restriction of the Killing form; on the compact form $-K|_{\mathfrak{m}}$ is positive definite.

Lemma 16 (Rank-Four Embedding) *Let the pull-back $g_{\mu\nu} = \mathfrak{g}_{\alpha\beta} \partial_\mu \phi^\alpha \partial_\nu \phi^\beta$ be non-degenerate at x . Then the differential $d\phi_x : T_x \mathbb{R}^4 \rightarrow \mathfrak{m}$ has rank four, and its image $\mathfrak{p}_0(x) = \Im d\phi_x$ is a four-dimensional subspace of $\mathfrak{m}$; the transverse 28 Goldstone directions span its $\mathfrak{g}$ -orthogonal complement $\mathfrak{p}_m(x)$.*

Proof Since $\dim T_x \mathbb{R}^4 = 4$, one has $\text{rank } d\phi_x \leq 4$, and because $\mathfrak{g}$ is positive definite $\text{rank } g_{\mu\nu} = \text{rank } d\phi_x$. Non-degeneracy of $g_{\mu\nu}$ therefore forces $\text{rank } d\phi_x = 4$, i.e. $d\phi_x$ is injective. $\square$

Choosing a $\mathfrak{g}$ -orthonormal basis of $\mathfrak{p}_0(x)$ and pulling back the Maurer-Cartan form gives the tetrad

$$e_\mu^a(x) = \theta_\alpha^a(\phi(x)) \partial_\mu \phi^\alpha(x), \quad a = 0, \dots, 3, \quad (2)$$

which by [Lemma 16](#) is invertible for a non-degenerate configuration. The induced (first fundamental) metric is

$$g_{\mu\nu}(x) = \delta_{ab} e_\mu^a(x) e_\nu^b(x), \quad (3)$$

where δ_{ab} is the restriction of the positive-definite $\mathfrak{g}$ to the orthonormal frame $\mathfrak{p}_0(x)$.

Theorem 17 (Emergent Riemannian Geometry) *For an open dense set of spacetime-filling configurations $\phi : \mathbb{R}^4 \rightarrow \mathcal{M}$, the pull-back (3) is a smooth, non-degenerate, positive-definite (Riemannian) metric on $\mathbb{R}^4$. It is the first fundamental form of the immersion ϕ into the Hermitian symmetric space $\mathcal{M}$, and no metric is postulated on $\mathbb{R}^4$ independently of ϕ .*

Proof Non-degeneracy and rank four are [Lemma 16](#); positive-definiteness is inherited from that of $\mathfrak{g}$ ([Theorem 22](#)), since $g_{\mu\nu} v^\mu v^\nu = \mathfrak{g}(d\phi_x v, d\phi_x v) \geq 0$ with equality iff $v = 0$. $\square$

The tetrad e_μ^a is thereby the fundamental gravitational field, and its dynamics follow from the σ -model action of [§ 4](#).

Remark 18 (Recovery of Lorentzian Spacetime) The coset metric $\mathfrak{g}$ is positive definite, so the induced geometry of [Theorem 17](#) is necessarily Riemannian: a pull-back of a definite form cannot have indefinite signature. This is the appropriate object, because the Schwinger functions of [§ 2](#) are Euclidean. Physical Lorentzian spacetime is recovered from those Euclidean correlation functions by Osterwalder–Schrader reconstruction [\[8\]](#), equivalently by Wick rotation of the reconstructed frame, exactly as in Euclidean quantum gravity.

Remark 19 (Emergent Spacetime and Rigidity) The image of $\mathbb{R}^4$ inside $\mathcal{M}$ is the physical spacetime manifold, and its metric (3) emerges from the field configuration rather than being put in by hand. The four frame directions $\mathfrak{p}_0(x)$ are not an H -invariant subspace: by Lemma 21 the isotropy representation on $\mathfrak{m}$ is irreducible, so $\mathfrak{m}$ admits no H -invariant four-plane. The splitting $\mathfrak{m} = \mathfrak{p}_0 \oplus \mathfrak{p}_m$ is therefore selected spontaneously by the configuration $\phi(x)$, in the same sense that a vacuum expectation value selects a direction; there is no contradiction between the emergence of a preferred frame and the H -irreducibility of the coset scalars.

With the vierbein $e_\mu^a(x)$ invertible and Theorem 17 establishing a non-degenerate induced metric (Lorentzian after continuation, Remark 18), $\mathbb{R}^4$ acquires the interpretation of physical spacetime. We now catalogue the dynamical fields that live on it and transform under the unbroken group $H = \text{Spin}(10) \times \text{U}(1)$.

Introduce the connection one-form

$$A_\mu(x) = A_\mu^A(x) T_A \in \mathfrak{h}, \quad (4)$$

which transforms by the affine shift $A_\mu \mapsto h A_\mu h^{-1} + h \partial_\mu h^{-1}$ for $h(x) \in H$. The adjoint of $\mathfrak{h}$ contains 45 generators of $\mathfrak{so}(10)$ and one of $\mathfrak{u}(1)$, hence $C_A = 46$ gauge bosons. The 45 $\text{Spin}(10)$ bosons are massless, each with two physical polarizations; the $\text{U}(1)_\psi$ boson acquires a Stückelberg mass once the overall phase mode is absorbed (Corollary 13, § 8).

The truncation contains a single left-handed Weyl multiplet in the chiral spinor $\mathbf{16}_{+1}$ of $\text{Spin}(10)$; all fermions are sections of the bundle $S \otimes \mathcal{V}_{\mathbf{16}}$, where S is the Weyl spin bundle determined by the emergent vierbein [12]. Thus $n_F = 16$ denotes the $\text{Spin}(10)$ multiplicity, corresponding to 32 real fermionic components, in the beta-function normalization used below.

The 32 coordinates of $\mathcal{M}$ form a single real-irreducible multiplet of $H = \text{Spin}(10) \times \text{U}(1)$ (Lemma 21), whose complexification is $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$. There is thus no H -invariant $4+28$ decomposition; the four frame directions $\mathfrak{p}_0$ singled out by Lemma 16 are selected spontaneously by the spacetime-filling configuration (Remark 19). Relative to a given background, the four longitudinal directions build the tetrad, while the 28 transverse fluctuations $\mathfrak{p}_m$ acquire a mass of order $f\mu$ (Theorem 26). All $n_S = 32$ real scalars enter the functional renormalization-group flow of § 5.

Table 1 Low-energy field content.

Sector	H representation	Spin	Real d.o.f.	Multiplicity n
Coset scalars	$\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$	0	32	n_S
Chiral fermions	$\mathbf{16}_{+1}$	$\frac{1}{2}$	32	n_F
Gauge bosons	adj_H	1	92	C_A

Remark 20 On a fixed non-degenerate background the tetrad e_μ^a maps the world-manifold index μ to the four-dimensional frame $\mathfrak{p}_0(x)$ selected by that configuration, and so acts as a soldering form between the emergent geometry and the coset. This soldering is a property of the configuration, not an H -invariant identification (Remark 19); it ensures that all matter fields couple consistently to the induced frame.

Collecting e_μ^a , A_μ , the chiral fermion ψ , and the coset scalars ϕ^α gives the field content used in the low-energy truncation.

4 Coset σ -Model, Quantization, and Induced Gravity

This section works in the coset representation of the observable net introduced in § 2. The σ -model with target $\mathcal{M}$ specifies the local fields whose bounded functional calculi generate the coset, gauge, and matter subalgebras. Its Euclidean path integral is used as a computational representation of the corresponding Schwinger functions. Within this representation we derive the one-loop Einstein–Hilbert term induced by matter fluctuations and identify the coset mass spectrum. The Planck scale is then set by the σ -model decay constant and the ultraviolet regularization.

Let $g(x) \in E_6$ be a local coset representative. The left-invariant Maurer-Cartan form is

$$\theta_\mu(x) = g^{-1}(x) \partial_\mu g(x) \in \mathfrak{e}_6. \quad (5)$$

Decomposing θ_μ into unbroken $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$ and broken $\mathfrak{m}$ components gives

$$\theta_\mu = A_\mu^i H_i + e_\mu^\alpha X_\alpha,$$

with $H_i \in \mathfrak{h}$, $X_\alpha \in \mathfrak{m}$, and coefficient functions $A_\mu^i(x)$ and $e_\mu^\alpha(x)$. α, β run over the 32 broken generators while i, j label the 45 generators of $\mathfrak{h}$.

Lemma 21 *The isotropy representation of $H = \text{Spin}(10) \times \text{U}(1)$ on the broken subspace $\mathfrak{m}$ is irreducible.*

Proof As an H -module, $\mathfrak{m}_\mathbb{C} \cong \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ [4, 5]. Each summand is irreducible, and the two are exchanged by complex conjugation. A proper nonzero H -invariant real subspace of $\mathfrak{m}$ would complexify to a conjugation-stable invariant subspace of $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$; the only invariant subspaces are 0, the two summands, and the whole space, and the summands are not conjugation-stable. Hence $\mathfrak{m}$ is irreducible as a real H -module. It is of complex type: its real commutant is $\mathbb{C} = \text{span}\{\mathbf{1}, J\}$, where J is the invariant complex structure of the Hermitian symmetric space EIII. $\square$

Theorem 22 *Up to an overall positive constant f^2 , there exists a unique E_6 -invariant symmetric bilinear form $\mathfrak{g}_{\alpha\beta}$ on $\mathfrak{m}$, namely $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$, where $K_{\alpha\beta}$ is the restriction to $\mathfrak{m}$ of the Killing form on $\mathfrak{e}_6$. On the compact real form of E_6 the Killing form is negative definite, so $\mathfrak{g}_{\alpha\beta}$ is positive definite.*

Proof See appendix for a short Schur-lemma argument. $\square$

By the coset construction of nonlinear realizations [13], the broken generators X_α furnish local coordinates $\phi^\alpha(x)$ on the vacuum manifold $\mathcal{M}$. The target-space metric $\mathfrak{g}_{\alpha\beta}$ is the unique E_6 -invariant metric (Theorem 22), and the spacetime metric $g_{\mu\nu}$ emerges from the vierbein pull-back construction of spacetime. We introduce the $\mathfrak{h}$ -valued gauge connection $A_\mu = A_\mu^i T_i$, and define

$$D_\mu \phi^\alpha = \partial_\mu \phi^\alpha + A_\mu^i (t_i)^\alpha{}_\beta \phi^\beta, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu],$$

where $(t_i)^\alpha{}_\beta$ are the representation matrices of $\mathfrak{h}$ on the coset coordinates.

By standard effective-field-theory arguments [13, 14], the low-energy action is organized as a derivative expansion constrained by the symmetries. In the minimal two-derivative truncation, with local H and Lorentz invariance imposed and with a single left-handed Weyl fermion $\psi(x)$ in the $\mathbf{16}_{+1}$ of $\text{Spin}(10)$, we use

$$S[\phi, A, \psi] = \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \mathfrak{g}_{\alpha\beta} D_\mu \phi^\alpha D_\nu \phi^\beta - \frac{1}{4} g^{-2} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + \bar{\psi} i \gamma^\mu D_\mu \psi - y \bar{\psi} \phi^\alpha T_\alpha \psi \right]. \quad (6)$$

The dimensionless couplings g and y , together with the decay constant f , depend on the renormalization scale; their flow is studied in § 5. Equation (6) is displayed in Lorentzian signature; the Euclidean functional integral (7) employs its Wick-rotated form S_E , obtained by continuation onto the Riemannian induced background of Theorem 17 ($\sqrt{-g} \rightarrow \sqrt{g}$, $e^{iS} \rightarrow e^{-S_E}$), in accordance with Remark 18. The Yukawa term is schematic: for a single $\mathbf{16}_{+1}$ there is no $\text{Spin}(10)$ -invariant cubic $\bar{\psi}\phi\psi$, since neither $\mathbf{16}$ nor $\overline{\mathbf{16}}$ occurs in $\mathbf{16} \otimes \mathbf{16} = \mathbf{10} \oplus \mathbf{120} \oplus \mathbf{126}$; it is defined only on completion of the matter to a full $\mathbf{27}$, or replaced by the standard derivative coupling of matter to the Goldstone current, and plays no role in the coset kinematics or the induced-gravity result below.

Lemma 23 *For $f^2 > 0$, the Euclidean continuation of the scalar kinetic term in (6) is positive definite.*

Proof Since $\mathfrak{g}_{\alpha\beta}$ is positive definite on the broken subspace, $D_\mu \phi^\alpha D^\mu \phi^\beta \mathfrak{g}_{\alpha\beta} \geq 0$, with equality only for vanishing covariant derivative. $\square$

The nonlinear sigma model action (6), continued to Euclidean signature, determines the Euclidean Schwinger functions of the coset representation. Gauge redundancy under local $\text{Spin}(10) \times \text{U}(1)$ is fixed by the Faddeev-Popov procedure [15], and the formal generating functional is

$$Z = \int \mathcal{D}\phi \mathcal{D}A \mathcal{D}\psi \exp[-S_E[\phi, A, \psi]]. \quad (7)$$

When these Schwinger functions satisfy reflection positivity, Euclidean invariance, symmetry, and clustering, Osterwalder–Schrader reconstruction recovers the Hilbert-space representation and local observable net described in § 2 [8, 9].

Expanding around a slowly varying background $X_0 \in \mathcal{M}$ and integrating out the quadratic fluctuations by Gaussian functional integration yields the one-loop effective action [16]

$$\Gamma^{(1)} = \frac{1}{2} \log \det(-\Delta_\phi) - \log \det(-\Delta_{\text{ghost}}) + \cdots, \quad (8)$$

where $-\Delta_\phi$ and $-\Delta_{\text{ghost}}$ are the second-variation operators for the physical and ghost fields, respectively. Its quadratic expansion in the fluctuation $\varphi = X - X_0 \in T_{X_0} \mathcal{M}$ reads

$$\Gamma^{(1)}[X_0 + \varphi] = \Gamma^{(1)}[X_0] + \frac{1}{2} \int d^4x \varphi^I [\text{Hess } V(X_0)]_{IJ} \varphi^J + \cdots, \quad (9)$$

so that the eigenvalues of the Hessian of the effective potential govern the one-loop mass spectrum (Theorem 26). The induced Einstein-Hilbert term and higher-curvature operators then follow from the standard heat-kernel expansion, as stated in Theorem 25.

Remark 24 We evaluate all functional determinants in Euclidean signature on a four-torus $\mathbb{T}^4$ of finite volume V and later take the thermodynamic limit $V \rightarrow \infty$; the final result is regulator-independent.

Theorem 25 (Induced Gravity) *One-loop quantum fluctuations induce an Einstein-Hilbert term*

$$\Gamma^{(1)} \supset \frac{1}{2} M_{\text{Pl}}^2 \int d^4x \sqrt{-g} R, \quad (10)$$

generated radiatively rather than postulated. The contribution of the $N_s = 32$ Goldstone scalars alone is

$$M_{\text{Pl}, \text{scalar}}^2 = \frac{N_s f^2}{(4\pi)^2} \log \frac{\Lambda^2}{\mu^2} > 0,$$

with Λ the cutoff and μ the renormalization scale. The full coefficient receives additional contributions from the Weyl and gauge sectors (Table C1), whose relative sign is scheme dependent; the invariant content of the theorem is the one-loop origin of the term, not the numerical value of M_{Pl}^2 .

Proof See Appendix C, which follows Sakharov’s induced-gravity argument and employs the heat-kernel coefficients listed in Table C1. The numerical coefficient in (10) is truncation- and scheme-dependent; the invariant content of the theorem is the one-loop origin of the term. The scalar-sector contribution is positive, while the sign of the full coefficient depends on the Weyl and gauge sectors and is scheme-dependent. $\square$

Theorem 26 (Mass Spectrum) *The one-loop effective potential is H -invariant, and $\mathfrak{m}$ is H -irreducible (Lemma 21); hence, by Schur’s lemma, its Hessian at the vacuum is a multiple of the identity,*

$$\text{Hess } V(X_0) = m^2 \mathbf{1}_{32}.$$

If the one-loop correction is stabilizing, so that $m^2 > 0$, the common (pseudo-Goldstone) mass is $m^2 = f^2 \mu^2$ in the normalization of (6). On a spacetime-filling background the four longitudinal directions $\mathfrak{p}_0$ of Lemma 16 are the tetrad (gravitational) degrees of freedom; the remaining 28 transverse fluctuations $\mathfrak{p}_m$ are physical scalars of common mass $m^2 = f^2 \mu^2$.

Proof See appendix. The tree potential of Theorem 11 is E_6 -invariant and vanishes on the entire orbit $\mathcal{M}$, so all 32 coset scalars are exact Goldstone bosons at tree level. The gauge and Yukawa couplings of (6) respect only H , so the one-loop effective potential is H -invariant but not E_6 -invariant and lifts these pseudo-Goldstones. Its Hessian commutes with H ; since $\mathfrak{m}$ is H -irreducible (Lemma 21), Schur's lemma forces $\text{Hess } V(X_0) = m^2 \mathbf{1}_{32}$ with a single $m^2 = f^2 \mu^2 > 0$ fixed by the flow. On a non-degenerate background the frame condensate $\langle \partial_\mu \phi^a \rangle = e_\mu^a$ reinterprets the four directions $\mathfrak{p}_0$ as the tetrad, promoted to the gravitational field by the induced Einstein-Hilbert term of Theorem 25; the 28 transverse fluctuations remain as physical scalars, each of mass $m^2 = f^2 \mu^2$. $\square$

The coefficient a_4 in the heat-kernel expansion generates

$$\alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \gamma R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}.$$

Theorem 27 *In the truncation of § 5, the dimensionless couplings α, β, γ scale as k^{-2} at large momentum scale k ; the corresponding operators are irrelevant within the effective coset flow below the cutoff $\epsilon < 1/6$.*

Proof Since $a_4 \propto f^0$ while $M_{\text{Pl}}^2 \propto f^2$, one has $\alpha \sim a_4/M_{\text{Pl}}^2 \propto k^{-2}$; the same reasoning applies to β and γ . $\square$

The construction contains a single dimensionful input: the Goldstone decay constant f that normalizes the invariant metric of Theorem 22. Every other scale is derived. The Planck mass is not postulated but generated radiatively by Theorem 25, with the one-loop enhancement factor $N_s \log(\Lambda^2/\mu^2)/(4\pi)^2$ placing f within roughly an order of magnitude of M_{Pl} itself. The grand-unified scale is the heterotic matching scale of § 5, fixed by the crossing of the running gauge couplings rather than by any parameter of the coset action. The electroweak scale enters only through the subsequent $\text{Spin}(10) \rightarrow \text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ descent (Corollary 65) and is not determined by the truncation studied here. The resulting hierarchy is:

Scale	Definition	Order of Magnitude
Goldstone decay constant	f	input
Planck mass	M_{Pl}	$\gtrsim 10^{18} \text{ GeV}$
Heterotic GUT matching scale	M_{GUT}	$\simeq 2 \times 10^{16} \text{ GeV}$
Electroweak VEV	v	246 GeV

The 28 transverse coset scalars of Theorem 26, with common mass of order $f\mu$, sit at the ultraviolet end of this hierarchy and communicate with the lower scales only through Planck-suppressed operators; their role in what follows is structural, as the source of the induced gravitational dynamics, rather than phenomenological.

5 Functional Renormalization-Group Flow

We now study the coset-field theory under continuous coarse-graining. After introducing the functional renormalization-group (RG) machinery and the dimensionless couplings, we compute the one- and two-loop β -functions in a finite truncation, identify the domain in which that truncation is valid, and determine the grand-unified matching scale.

We work with the Wetterich equation

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr}[(\Gamma_k^{(2)} + R_k)^{-1} \partial_k R_k],$$

choosing Litim's regulator $R_k(p^2) = Z_k(k^2 - p^2)\theta(k^2 - p^2)$, which optimizes threshold integrals while preserving locality. The natural dimensionless couplings are

$$\xi = \frac{k^2}{f^2}, \quad \hat{g}^2 = \frac{k^2 g^2}{f^2}, \quad \hat{y}^2 = \frac{k^2 y^2}{f^2}, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2}.$$

We also introduce the coset loop-expansion parameter

$$\epsilon \equiv \frac{\xi}{16\pi^2} = \frac{k^2}{16\pi^2 f^2},$$

which controls the derivative expansion of the nonlinear σ -model; the validity boundary used throughout the paper is stated in terms of this parameter ([Remark 29](#)). Projecting the flow onto the coefficients of $\sqrt{-g} R$, $\sqrt{-g} \bar{\psi} \gamma^\mu D_\mu \psi$, and analogous invariants yields the β -functions quoted below.

Remark 28 We evaluate functional determinants on a finite four-torus $\mathbb{T}^4$ and take $V_{\mathbb{T}^4} \rightarrow \infty$ at the end, removing all infrared regulators without affecting ultraviolet behavior.

Remark 29 (Validity Boundary) The coset RG equations in this section are used only in the effective regime

$$\epsilon < \frac{1}{6}.$$

At $\epsilon = 1/6$, the local coset description reaches the matching boundary where the $E_8 \times E_8$ heterotic string background supplies the ultraviolet completion. Consequently, a UV-repulsive direction at this boundary is interpreted as a signal that the effective coset variables are no longer the correct coordinates beyond the cutoff, provided the heterotic matching conditions are satisfied.

At one loop, the running of the gauge, scalar, and gravitational couplings is determined by the field content listed in [Table 1](#). Working in the minimal-subtraction scheme (or, equivalently, in the functional RG, with a Litim regulator), the standard group-theoretic result for a gauge algebra $\mathfrak{g}$ with n_F Weyl fermions and n_S real scalars

in the fundamental representation reads [17]

$$\beta_{g^{-2}} = \frac{b_0}{48\pi^2}, \quad b_0 = \frac{11}{3} C_A - \frac{4}{3} T_R n_F - \frac{1}{6} T_R n_S.$$

Within E_6 , we have $C_A = 12$, $T_R = 1$, and matter multiplicities $(n_F, n_S) = (16, 32)$, yielding $b_0 = 52/3$. These coefficients are convention- and scheme-dependent book-keeping for the printed reduced β -system: the value mixes the E_6 normalization $C_A = 12$ with a unit matter index $T_R = 1$, and the coupling that runs below the coset scale is that of the unbroken $H = \text{Spin}(10) \times U(1)$ rather than E_6 . Only the qualitative structure of the flow—not the numerical value or a convention-independent sign of b_0 —enters what follows.

Throughout the paper, we trade the dimensionless gauge coupling $g^2(k)$ for the canonically rescaled

$$\hat{g}^2(k) = \frac{k^2}{f^2} g^2(k),$$

so that $\hat{g}^2$ has engineering dimension 2. Using $\beta_{\hat{g}^2} = 2\hat{g}^2 \beta_g/g$, we find

$$\beta_{\hat{g}^2} = -\frac{b_0}{24\pi^2} \hat{g}^4.$$

The coset-field wave-function f^2 is protected by the E_6 symmetry at this order, hence $\beta_{f^2} = 0$. The dimensionful ratio $\xi(k) = k^2/f^2$ therefore runs only through its canonical dimension: $\beta_\xi = 2\xi$.

Finally, integrating out the 28 massive coset fields generates the one-loop Sakharov flow of the induced Planck mass [6, 7]:²

$$\beta_\kappa = 2\kappa + \frac{5}{48\pi^2} \kappa^2, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2(k)}.$$

Collecting the three directions, we have

$$\beta_{\hat{g}^2} = -\frac{b_0}{24\pi^2} \hat{g}^4, \quad \beta_\xi = 2\xi, \quad \beta_\kappa = 2\kappa + \frac{5}{48\pi^2} \kappa^2. \quad (11)$$

Coset-scalar self-interactions and Yukawa diagrams enter at two loops. Keeping the leading structures in the truncation gives

$$\begin{aligned} \beta_{\hat{g}^2} &= -\frac{b_0}{24\pi^2} \hat{g}^4 - \frac{b_1}{(16\pi^2)^2} \hat{g}^6 + \frac{c_1}{16\pi^2} \hat{g}^4 \hat{y}^2, \\ \beta_{\hat{y}^2} &= b_y \hat{y}^4 + c_2 \hat{g}^2 \hat{y}^2, \\ \beta_\xi &= 2\xi + A \hat{y}^2 \xi + b_0 \hat{g}^2 \xi. \end{aligned} \quad (12)$$

²The minus sign follows from $\mu \partial_\mu \log(\Lambda^2/\mu^2) = -2$.

The coefficients b_1 , c_1 , c_2 , b_y , and A are fixed by the field content of [Table 1](#); their values, together with the projection rules used to extract them from the Wetterich equation, are derived in [Appendix D](#) and collected in [Table D2](#). Two sign facts control everything that follows: $b_0, b_1 > 0$, so both pure-gauge terms drive $\hat{g}^2$ toward asymptotic freedom, while $c_1 < 0$, so the Yukawa correction cannot balance them at positive coupling. These signs are the entire input to the fixed-point analysis of the next subsection.

Theorem 30 (No Positive-Coupling Non-Gaussian Coset Fixed Point) *For the β -system (12) with the coefficients of [Table D2](#), the only nontrivial real gauge–Yukawa branch has $\hat{g}_*^2 < 0$ and is therefore outside the physical positive-coupling domain. The physical coset flow below the cutoff $\epsilon < 1/6$ is consequently an effective flow, not an interacting asymptotic-safety completion. At the cutoff boundary the ultraviolet description is supplied by the $E_8 \times E_8$ heterotic theory.*

Proof Insert the coefficients of [Table D2](#) into (12). For $\hat{g}^2 > 0$ and $\hat{y}^2 > 0$, every term in

$$\beta_{\hat{g}^2} = -\frac{b_0}{24\pi^2}\hat{g}^4 - \frac{b_1}{(16\pi^2)^2}\hat{g}^6 + \frac{c_1}{16\pi^2}\hat{g}^4\hat{y}^2$$

is strictly negative, since $b_0, b_1 > 0$ and $c_1 < 0$. Hence $\beta_{\hat{g}^2}$ cannot vanish in the positive-coupling interior. Solving the algebraic system gives the nontrivial real branch

$$\hat{g}_*^2 \simeq -0.567, \quad \hat{y}_*^2 \simeq -1.70,$$

which is not a physical coupling branch. The κ -flow

$$\beta_\kappa = 2\kappa + \frac{5}{48\pi^2}\kappa^2$$

has its nontrivial zero at negative κ , while the positive- κ flow is repulsive toward the heterotic matching boundary. This is consistent with [Remark 29](#): the coset truncation is not required to provide its own interacting ultraviolet fixed point beyond $\epsilon = 1/6$. $\square$

Running the gauge couplings downward, M_{GUT} is defined by the crossing $\alpha_2^{-1}(k) = \alpha_1^{-1}(k)$. Using supersymmetric/heterotic-threshold one-loop slopes and standard electroweak inputs at M_Z , $M_{\text{GUT}} \simeq 2 \times 10^{16}$ GeV, with a 5% uncertainty from two-loop and threshold effects. With the plain non-supersymmetric Standard-Model one-loop coefficients $b_1 = 41/10$ and $b_2 = -19/6$, the same crossing occurs near 10^{13} GeV. Thus the scale quoted here is a heterotic matching scale, not a prediction of the nonsupersymmetric coset truncation alone.

Lemma 31 *Curvature-squared and higher coset-field operators are irrelevant at the Gaussian effective branch within the derivative expansion considered here: $\Delta < 0$ for the retained higher operators.*

Proof Each operator carries an explicit k^{-2} relative to the Einstein-Hilbert term. With $M_{\text{Pl}}(k)$ generated by the induced Sakharov flow, one finds the canonical suppression $\Delta = -2 + \mathcal{O}(\kappa)$ throughout the effective regime below $\epsilon < 1/6$. $\square$

Lemma 32 *The dimensionless coupling $\kappa = k^2/M_{\text{Pl}}^2$ has a Gaussian weak-coupling branch in the induced-gravity truncation, while the nontrivial zero lies at negative κ and is not a physical UV fixed point.*

Proof Solving β_κ on the Gaussian branch yields $\kappa(k) = 2\pi^2[\pi^2 + 5\log(k/\Lambda)]^{-1} \rightarrow 0$. The other zero is $\kappa_* = -96\pi^2/5 < 0$, outside the positive induced-gravity domain. $\square$

Corollary 33 (Metastability Estimate) *Under the semiclassical bounce estimate used in the truncation, the vacuum is metastable with lifetime exceeding 10^{10} yr.*

Proof The bounce action satisfies $S_B \sim M_{\text{Pl}}^4/\alpha \gtrsim 400$, so $\Gamma/V \propto e^{-S_B}$ is well below observational limits. $\square$

Theorem 34 (Heterotic Ultraviolet Criterion) *In the truncation defined by the coset-field action (6), the positive-coupling two-loop flow does not furnish an interacting asymptotic-safety fixed point. The coset theory is instead an effective theory below $\epsilon < 1/6$. If an $E_8 \times E_8$ heterotic string background satisfies the matching conditions of § 7, then it supplies the ultraviolet completion at the boundary.*

Proof Theorem 30 shows that the nontrivial branch of the printed two-loop equations is outside the positive-coupling domain. Remark 29 specifies the cutoff at which the effective coset description must be matched to a UV theory. The local matching data and global consistency conditions are stated in § 7. When those conditions hold, the completion is heterotic rather than an autonomous non-Gaussian coset fixed point. $\square$

6 The Coset Spectral Zeta Sector

The zeta function enters UCFT through a commutative spectral subalgebra of the coset observable algebra. The object on the UCFT side is the nonnegative self-adjoint coset mass-squared operator associated with an EIII vacuum. The arithmetic input is the shifted divisor

$$m_\rho^2 = -\left(\rho - \frac{1}{2}\right)^2$$

attached to the nontrivial zeros of the completed Riemann zeta function. The coset spectral zeta sector is the assertion that this arithmetic divisor is realized, equivariantly and with the mass-squared coordinate preserved, inside the positive spectrum of that coset subalgebra.

This section makes that assertion precise and derives its consequence. The finite adelic orbit data explain why rational primes supply the arithmetic trace. The completed zeta factor supplies the archimedean normalization. The coset spectral operator supplies positivity. Once these three pieces are identified by a Galois-equivariant coset spectral realization, the shifted zeta divisor is forced to lie in $\mathbb{R}_{\geq 0}$, which is equivalent to the Riemann hypothesis.

Let $\mathbb{B} = \{0, 1\}$ and define

$$\mathcal{E} : \mathbb{C} \setminus \mathbb{B} \longrightarrow \mathbb{C}, \quad \mathcal{E}(s) = \frac{1}{2} \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Equivalently, if

$$\xi_{\mathbb{R}}(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

denotes Riemann's completed entire zeta function [18–20], then $\mathcal{E}(s) = \xi_{\mathbb{R}}(s)/(s(s-1))$. Set

$$\lambda = \left(s - \frac{1}{2}\right)^2, \quad \Phi(\lambda) = \mathcal{E}\left(\frac{1}{2} + \sqrt{\lambda}\right).$$

The functional equation gives $\mathcal{E}(\frac{1}{2} + z) = \mathcal{E}(\frac{1}{2} - z)$, so Φ is independent of the branch of $\sqrt{\lambda}$.

The coordinate λ is the bridge to the coset sector. On the UCFT side it is interpreted as the negative of a mass-squared eigenvalue. Thus the question of whether zeros of $\mathcal{E}$ lie on the critical line becomes the question of whether the corresponding shifted mass-squared values are real and nonnegative in the coset spectrum. The definitions below make this spectral identification precise.

Definition 35 (Finite Adelic Albert–EIII Completion) Let $J_{\mathbb{Z}}$ be an integral form of the Albert algebra, the *integral Albert–EIII charge lattice*; for definiteness one may take the lattice of Hermitian 3×3 matrices over the Coxeter order of integral octonions. Its finite adelic completion is

$$J_{\mathbb{A}_f} = J_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{A}_f.$$

Definition 36 (Zeta Sector) The *zeta sector* $\mathcal{H}_{\zeta} \subset \mathcal{H}$ is the closed Hilbert subspace selected, under the coset correspondence, by $J_{\mathbb{A}_f}$ together with a one-dimensional automorphic character (the *zeta automorphic character*).

Remark 37 The sector is normalized so that its finite orbit space is

$$\mathcal{O}_{\zeta} \cong \mathbb{A}_f^{\times} / \widehat{\mathbb{Z}}^{\times}, \quad \widehat{\mathbb{Z}}^{\times} = \prod_p \mathbb{Z}_p^{\times},$$

the idele-class datum of Tate's thesis [21].

Remark 38 (Archimedean Zeta Oscillator) The corresponding archimedean oscillator is

$$Z_{\infty}(s) = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right).$$

Theorem 39 (Zeta-Sector Quadratic Operator) *Suppose the zeta-sector quadratic form obtained from the second variation of the sectoral action is symmetric, closed, and semi-bounded. Then it defines a self-adjoint mass-squared operator*

$$M_{\zeta}^2 : \text{Dom}(M_{\zeta}^2) \subset \mathcal{H}_{\zeta} \longrightarrow \mathcal{H}_{\zeta}.$$

Proof This is the first representation theorem for closed, densely defined, semibounded quadratic forms (Kato), which specializes to the Friedrichs extension when the form is that of a semibounded symmetric operator. $\square$

Definition 40 (Zeta Determinant Realization) The zeta sector is said to realize the completed zeta determinant if there exists a nonzero constant $C \in \mathbb{C}^\times$ such that

$$\Phi(\lambda) = C \det_{\text{reg}} \left(\lambda I + M_\zeta^2 \right).$$

This is the spectral-determinant assumption in the sense of the trace-formula programs of Connes and Berry–Keating [22, 23], after passage to the shifted-square coordinate $\lambda = (s - \frac{1}{2})^2$.

Lemma 41 (Prime-Orbit Correspondence) *Call a finite zeta orbit primitive if it corresponds to an indecomposable element of the positive cone of $\mathcal{O}_\zeta \cong \bigoplus_p \mathbb{Z}$. There is a canonical bijection between primitive finite zeta orbits and rational primes.*

Proof Every finite idele $a = (a_p)_p \in \mathbb{A}_f^\times$ has a unique factorization [21]

$$a_p = p^{v_p(a_p)} u_p, \quad u_p \in \mathbb{Z}_p^\times,$$

where $v_p(a_p) \in \mathbb{Z}$ and $v_p(a_p) = 0$ for all but finitely many primes p . Hence

$$\mathbb{A}_f^\times \cong \left(\bigoplus_p p^\mathbb{Z} \right) \times \widehat{\mathbb{Z}}^\times, \quad \mathbb{A}_f^\times / \widehat{\mathbb{Z}}^\times \cong \bigoplus_p \mathbb{Z}.$$

The positive cone $\bigoplus_p \mathbb{Z}_{\geq 0}$ is the free commutative monoid of nonzero ideals of $\mathbb{Z}$, and its indecomposable elements are the prime ideals (p) . Primitive zeta orbits correspond to these indecomposable elements. $\square$

Corollary 42 (Prime-Orbit Length) *Under the idele-norm character, a primitive finite zeta orbit associated with the prime p has logarithmic length $\log p$.*

Lemma 43 (Finite Arithmetic Trace) *Define the finite prime-orbit trace as the orbit sum, in the sense of the trace-formula programs [22],*

$$\text{Tr}_{\text{fin}}(s) = - \sum_p \sum_{k \geq 1} (\log p) e^{-ks \log p} = - \sum_p \sum_{k \geq 1} (\log p) p^{-ks},$$

the sum running over the primitive orbits of Lemma 41 and their iterates, weighted by the lengths of Corollary 42. Then, for $\Re(s) > 1$,

$$\text{Tr}_{\text{fin}}(s) = \frac{\zeta'(s)}{\zeta(s)}.$$

Proof Equivalently,

$$\mathrm{Tr}_{\mathrm{fin}}(s) = - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s},$$

where Λ is the von Mangoldt function [24]. The Euler product for ζ , valid for $\Re(s) > 1$, gives the stated identity. $\square$

Lemma 44 (Completed zeta trace) *The logarithmic derivative of $\mathcal{E}$ is*

$$\frac{\mathcal{E}'(s)}{\mathcal{E}(s)} = -\frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{s}{2}\right) + \frac{\zeta'(s)}{\zeta(s)}, \quad \psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}.$$

Proof Take the logarithmic derivative of

$$\mathcal{E}(s) = \frac{1}{2} \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

$\square$

Putting $s = \frac{1}{2} + \sqrt{\lambda}$, Lemma 44 gives

$$\frac{\Phi'(\lambda)}{\Phi(\lambda)} = \frac{1}{2\sqrt{\lambda}} \left[-\frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{1}{4} + \frac{1}{2} \sqrt{\lambda}\right) + \frac{\zeta'(\frac{1}{2} + \sqrt{\lambda})}{\zeta(\frac{1}{2} + \sqrt{\lambda})} \right].$$

For $\Re(\frac{1}{2} + \sqrt{\lambda}) > 1$, this becomes

$$\frac{\Phi'(\lambda)}{\Phi(\lambda)} = \frac{1}{2\sqrt{\lambda}} \left[-\frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{1}{4} + \frac{1}{2} \sqrt{\lambda}\right) - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2+\sqrt{\lambda}}} \right].$$

Lemma 45 (Shifted-Square Equivalence) *The Riemann hypothesis is equivalent to the assertion that every zero of Φ lies on $\mathbb{R}_{\leq 0}$.*

Proof If the Riemann hypothesis holds, every zero of $\mathcal{E}$ has the form

$$s = \frac{1}{2} + i\gamma, \quad \gamma \in \mathbb{R}.$$

Thus

$$\lambda = \left(s - \frac{1}{2}\right)^2 = -\gamma^2 \leq 0.$$

Conversely, suppose every zero of Φ lies on $\mathbb{R}_{\leq 0}$. If ρ is a zero of $\mathcal{E}$, then

$$\lambda_{\rho} = \left(\rho - \frac{1}{2}\right)^2 = -\gamma^2$$

for some $\gamma \in \mathbb{R}$. Hence $\rho = \frac{1}{2} \pm i\gamma$, so $\Re(\rho) = \frac{1}{2}$. Since the zeros of $\mathcal{E}$ are precisely the nontrivial zeros of $\zeta(s)$, the Riemann hypothesis follows [20]. $\square$

Theorem 46 (Riemann Hypothesis as Coset Positivity) *Assume the Albert–EIII postulates and suppose the zeta divisor is realized by the positive UCFT coset spectral data in the sense of Definition 50. Then the Riemann hypothesis holds. Equivalently, the determinant has no tachyonic or complex mass-squared zeros: all determinant zeros occur at*

$$\lambda = -m_n^2, \quad m_n^2 \in \mathbb{R}_{\geq 0}.$$

This statement is a conditional reduction, not an existence theorem: the realization hypothesis is precisely the unconstructed Hilbert–Pólya datum (Remark 51), and nothing in this paper constructs it.

Proof The UCFT coset spectral operator of Definition 48 is self-adjoint and nonnegative after gauge fixing and removal of zero modes. Therefore its mass-squared spectrum lies in $\mathbb{R}_{\geq 0}$. A mass-coordinate-preserving realization of the zeta divisor in this coset spectrum forces every arithmetic value

$$m_\rho^2 = -\left(\rho - \frac{1}{2}\right)^2$$

to be real and nonnegative. Thus every zero of Φ lies on $\mathbb{R}_{\leq 0}$ in the shifted-square coordinate. By Lemma 45, this is equivalent to the Riemann hypothesis. In determinant language, the same statement says that zeros occur only at $\lambda = -m_n^2$ with $m_n^2 \geq 0$, so no tachyonic or complex mass-squared zeros occur. $\square$

Definition 47 (Arithmetic Mass Divisor) Let $\xi(s)$ denote the completed Riemann zeta function, and let

$$Z_\xi := \{\rho \in \mathbb{C} : \xi(\rho) = 0, \ 0 < \Re(\rho) < 1\}$$

be the nontrivial zero divisor, counted with multiplicity. Define the shifted arithmetic mass-squared divisor

$$\Sigma_\zeta := \left\{ m_\rho^2 := -\left(\rho - \frac{1}{2}\right)^2 : \rho \in Z_\xi \right\}.$$

The functional equation identifies the pair ρ and $1 - \rho$, since

$$-\left(1 - \rho - \frac{1}{2}\right)^2 = -\left(\rho - \frac{1}{2}\right)^2.$$

Complex conjugation acts by

$$m_\rho^2 \mapsto \overline{m_\rho^2}.$$

We equip Σ_ζ with its natural locally finite divisor topology.

Definition 48 (UCFT Spectral Algebra) Let $\phi_0 \in \text{EIII}$ be a vacuum and let

$$\Delta_{\text{UCFT}} = \nabla^* \nabla + \mathcal{R} + \text{Hess}_{\phi_0}(V)$$

be the quadratic coset operator in UCFT, realized in the zeta sector $\mathcal{H}_\zeta$ of Definition 36, so that $\Delta_{\text{UCFT}} = M_\zeta^2$ in the notation of Theorem 39. After the required gauge fixing and removal of zero modes, take Δ_{UCFT} to be a self-adjoint nonnegative operator on a Hilbert space $\mathcal{H}_{\text{UCFT}} \subseteq \mathcal{H}_\zeta$. Let

$$\Sigma_{\text{UCFT}} := \text{Spec}(\Delta_{\text{UCFT}}) \subseteq \mathbb{R}_{\geq 0}.$$

Define the commutative spectral C^* -algebra

$$\mathcal{A}_{\text{UCFT}} := C_0(\Sigma_{\text{UCFT}}).$$

Similarly define

$$\mathcal{A}_\zeta := C_0(\Sigma_\zeta),$$

where Σ_ζ is the arithmetic mass-squared divisor.

Remark 49 (Status of the Realization Hypothesis) The realization posited below is a hypothesis, not a construction. It is a statement about the sectoral operator $\Delta_{\text{UCFT}} = M_\zeta^2$ of the adelically selected zeta sector, not about the Laplace-type operator of the compact σ -model sector, whose spectrum is fixed by harmonic analysis on EIII and is manifestly not the arithmetic divisor. Nothing in this paper constructs a zeta sector whose spectrum is Σ_ζ ; the theorems below assert only the implication from its existence.

Definition 50 (Galois-Equivariant Spectral Realization) A Galois-equivariant spectral realization of the zeta data in the UCFT coset sector is a $*$ -isomorphism of commutative spectral C^* -algebras

$$\Theta : \mathcal{A}_\zeta \xrightarrow{\sim} \mathcal{A}_{\text{UCFT}}$$

such that:

1. Θ preserves the distinguished mass-squared coordinate;
2. Θ intertwines complex conjugation on Σ_ζ with the real structure of the UCFT Hilbert-space representation;
3. at every finite algebraic approximation level, Θ is compatible with the relevant Galois action on the arithmetic mass divisor and the induced algebraic symmetry action on the UCFT spectral data.

Remark 51 (Relation to the Hilbert–Pólya Conjecture) The construction above is a coset-sector instance of the Hilbert–Pólya conjecture: the proposal that the ordinates γ of the nontrivial zeros $\rho = \frac{1}{2} + i\gamma$ are the spectrum of a self-adjoint operator, so that self-adjointness alone forces $\gamma \in \mathbb{R}$ and hence the Riemann hypothesis [22, 25]. In our language, self-adjointness and nonnegativity of Δ_{UCFT} are free: they follow from the coset construction (Theorem 39) and require no further input. The entire content of the conjecture, in any of its forms, is naturality: the operator must arise independently of the zeros it is meant to reproduce, with a spectrum computable on its own terms. Absent that requirement the conjecture is nearly vacuous, since if the Riemann hypothesis holds then $\text{diag}(\gamma_n)$ on ℓ^2 is trivially self-adjoint with the correct spectrum. The Galois-equivariant realization Θ above is precisely a naturality condition of this kind: it demands that the isomorphism respect the arithmetic Galois action and the coset real structure simultaneously, not merely that some isomorphism of sets exists. Definition 36 and Lemma 41 supply the coset-side data meant to play the role of an independently computable trace formula, but identifying that data with the actual spectrum of Δ_{UCFT} , rather than positing it. We therefore regard the theorem below as a precise operator-algebraic form of Hilbert–Pólya naturality, sufficient for the Riemann hypothesis but not asserted to be necessary for it: a coset-external proof of the Riemann hypothesis would not contradict anything proved here.

Theorem 52 (Coset Spectral Criterion for the Riemann Hypothesis) *Assume the Albert–EIII postulates. If there exists a Galois-equivariant spectral realization*

$$\Theta : \mathcal{A}_\zeta \xrightarrow{\sim} \mathcal{A}_{\text{UCFT}},$$

then the Riemann hypothesis holds. This is a conditional reduction in the Hilbert–Pólya sense: the existence of Θ is the open hypothesis, and the theorem does not assert it.

Proof Since

$$\Theta : \mathcal{A}_\zeta \xrightarrow{\sim} \mathcal{A}_{\text{UCFT}}$$

is a $*$ -isomorphism of commutative C^* -algebras, Gelfand duality gives a homeomorphism of spectra

$$\Theta^\vee : \Sigma_{\text{UCFT}} \xrightarrow{\sim} \Sigma_\zeta.$$

The UCFT coset spectral operator is self-adjoint and nonnegative, hence

$$\Sigma_{\text{UCFT}} \subseteq \mathbb{R}_{\geq 0}.$$

Because the realization preserves the distinguished mass-squared coordinate, the transported arithmetic mass-squared divisor also satisfies

$$\Sigma_\zeta \subseteq \mathbb{R}_{\geq 0}.$$

Let $\rho \in Z_\xi$. Write

$$\rho = \frac{1}{2} + \alpha + i\gamma, \quad \alpha, \gamma \in \mathbb{R}.$$

Then

$$m_\rho^2 = -\left(\rho - \frac{1}{2}\right)^2 = -(\alpha + i\gamma)^2 = \gamma^2 - \alpha^2 - 2i\alpha\gamma.$$

Since $m_\rho^2 \in \mathbb{R}_{\geq 0}$, its imaginary part vanishes:

$$-2\alpha\gamma = 0.$$

For a nontrivial zero of the completed Riemann zeta function in the critical strip, $\gamma \neq 0$. Hence

$$\alpha = 0.$$

Thus

$$\Re(\rho) = \frac{1}{2}.$$

Since this holds for every nontrivial zero ρ , the Riemann hypothesis follows. $\square$

The criterion of [Theorem 52](#) is rigid at its endpoints: it demands a single $*$ -isomorphism identifying the arithmetic divisor with the coset spectrum exactly. For approximation arguments—in particular for any attempt to build a realization level by level from the finite adelic data of [Definition 36](#)—a deformation-adapted formulation is preferable. We therefore replace the single isomorphism by a continuous family of operators interpolating between the coset endpoint and the arithmetic endpoint.

Definition 53 (Locally Finite Divisor Topology) Let $\text{Div}_{\text{loc}}(\mathbb{C})$ denote the space of locally finite effective divisors on $\mathbb{C}$. Thus an element is a formal sum

$$D = \sum_{\lambda \in \mathbb{C}} m_\lambda [\lambda], \quad m_\lambda \in \mathbb{Z}_{\geq 0},$$

such that only finitely many λ 's occur in each compact subset of $\mathbb{C}$.

We equip $\text{Div}_{\text{loc}}(\mathbb{C})$ with the vague topology: a sequence D_n converges to D if and only if, for every compactly supported continuous test function $\varphi \in C_c(\mathbb{C})$,

$$\sum_{\lambda} m_{\lambda}^{(n)} \varphi(\lambda) \longrightarrow \sum_{\lambda} m_{\lambda} \varphi(\lambda).$$

Definition 54 (Spectral Continuity) Let B be a parameter space containing a coset point b_{UCFT} and an arithmetic point b_ζ . A spectral bridge for the coset correspondence is a family of closed operators

$$\{\mathcal{M}_b^2\}_{b \in B}$$

such that:

1. each $\mathcal{M}_b^2$ acts on a Hilbert space $\mathcal{H}_b$, and the $\mathcal{H}_b$ are unitarily identified with a fixed Hilbert space (or form a continuous field of Hilbert spaces over B), so that the resolvent comparison in the next item is meaningful;
2. the family is continuous in the norm-resolvent topology, meaning that for some, equivalently every, $w \in \mathbb{C} \setminus \mathbb{R}$,

$$b \mapsto (\mathcal{M}_b^2 - w)^{-1}$$

is norm-continuous;

3. each $\mathcal{M}_b^2$ has compact resolvent and zeta-regularizable spectrum, and hence admits a regularized determinant in the mass-squared coordinate,

$$D_b(\mu) := \det_{\text{reg}}(\mathcal{M}_b^2 - \mu),$$

whose zero divisor $\text{div}(D_b)$ is the eigenvalue divisor of $\mathcal{M}_b^2$, counted with multiplicity;

4. the zero-divisor map

$$b \mapsto \text{div}(D_b)$$

is continuous as a map

$$B \longrightarrow \text{Div}_{\text{loc}}(\mathbb{C}).$$

Definition 55 (Galois-Compatible Spectral Bridge) A spectral bridge is Galois-compatible if, at every finite algebraic approximation level, the divisor

$$\text{div}(D_b)$$

is stable under the relevant Galois action, and the transition maps in the bridge intertwine this Galois action with the real structure on the quantum Hilbert spaces $\mathcal{H}_b$.

Equivalently, for every admissible finite approximation $D_{b,N}$, every Galois element σ , and every point λ of the divisor, one has

$$\sigma(\lambda) \in \text{supp}(D_{b,N})$$

with the same multiplicity, and this compatibility is preserved under passage to the limiting divisor.

Definition 56 (Positive Continuous Albert–Zeta Realization) A *Positive Continuous Albert–Zeta Realization* is a Galois-compatible spectral bridge

$$\{\mathcal{M}_b^2\}_{b \in B}$$

such that:

1. the UCFT endpoint satisfies

$$\mathcal{M}_{b_{\text{UCFT}}}^2 = \Delta_{\text{UCFT}};$$

2. the arithmetic endpoint satisfies

$$\text{div}(D_{b_\zeta}) = \Sigma_\zeta = \left\{ -\left(\rho - \frac{1}{2}\right)^2 : \xi(\rho) = 0, 0 < \Re(\rho) < 1 \right\};$$

3. every operator in the bridge is self-adjoint and nonnegative:

$$\mathcal{M}_b^2 = \left(\mathcal{M}_b^2\right)^*, \quad \mathcal{M}_b^2 \geq 0$$

for all $b \in B$.

Theorem 57 (Continuous Coset Criterion for the Riemann Hypothesis) *Assume the Albert–EIII postulates. If a Positive Continuous Albert–Zeta Realization exists, then the Riemann hypothesis holds. As with Theorem 52, this is a conditional reduction: the existence of the realization is the open hypothesis.*

Proof For each $b \in B$, self-adjointness implies

$$\text{Spec}(\mathcal{M}_b^2) \subseteq \mathbb{R}.$$

Nonnegativity strengthens this to

$$\text{Spec}(\mathcal{M}_b^2) \subseteq \mathbb{R}_{\geq 0}.$$

Since

$$D_b(\mu) = \det_{\text{reg}}(\mathcal{M}_b^2 - \mu),$$

the zero divisor of D_b lies in the mass-squared spectrum:

$$\text{div}(D_b) \subseteq \mathbb{R}_{\geq 0}.$$

At the arithmetic endpoint $b = b_\zeta$, the zero divisor is

$$\Sigma_\zeta = \left\{ -\left(\rho - \frac{1}{2}\right)^2 : \xi(\rho) = 0, 0 < \Re(\rho) < 1 \right\}.$$

Thus every shifted arithmetic mass-squared value is real and nonnegative.

Let

$$\rho = \frac{1}{2} + \alpha + i\gamma, \quad \alpha, \gamma \in \mathbb{R}.$$

Then

$$-\left(\rho - \frac{1}{2}\right)^2 = -(\alpha + i\gamma)^2 = \gamma^2 - \alpha^2 - 2i\alpha\gamma.$$

Since this number is real, its imaginary part vanishes:

$$-2\alpha\gamma = 0.$$

A nontrivial zero of $\xi(s)$ in the critical strip is not real, so

$$\gamma \neq 0.$$

Hence

$$\alpha = 0.$$

Therefore

$$\Re(\rho) = \frac{1}{2}.$$

Since this holds for every nontrivial zero, the Riemann hypothesis follows. $\square$

7 Heterotic Correspondence and the Albert–EIII Swampland Criterion

The construction developed above realizes gravity and spacetime as induced structures: there is no fundamental graviton, and the metric $g_{\mu\nu}$ is the pull-back of the EIII metric along a spacetime-filling configuration (Theorem 17), while the Planck mass is generated radiatively in the sense of Sakharov and Adler [6, 7]. This motivates a precise, but limited, comparison between the coset sector and the massless bosonic sector of an $E_8 \times E_8$ heterotic compactification [26] whose visible gauge group is E_6 . We emphasize at the outset that the statement established here is local, two-derivative, and current-algebraic. The purpose of this section is to formulate the heterotic correspondence, identify a concrete $\mathbb{Z}_{12}$ heterotic candidate, and state the resulting Albert–EIII swampland criterion.

Fix an $E_8 \times E_8$ heterotic background on $\mathbb{R}^{1,3} \times X$ with X a six-dimensional internal space whose worldsheet realization preserves an E_6 current algebra at level one, so that the visible massless spectrum organizes into E_6 multiplets and the **78** adjoint is present as a gauge multiplet. Let M_{res} denote the residual scalar target obtained after the internal CFT has fixed all moduli except those of the visible E_6 -breaking sector. We write $H = \text{Spin}(10) \times \text{U}(1)_\psi$ for the grand-unified isotropy group singled out by an adjoint Higgs vacuum in the $E_6 \supset H$ singlet direction (Theorem 63).

Definition 58 (Heterotic Correspondence Map) The *heterotic correspondence map* is the E_6 -equivariant map

$$F : M_{\text{res}} \longrightarrow \mathcal{M} = E_6 / (\text{Spin}(10) \times \text{U}(1))$$

sending the gauge-unfixed adjoint-Higgs orbit of the heterotic model onto $\mathcal{M}$, and identifying the residual scalar fluctuations with the Goldstone coordinates ϕ^α of § 3.

Lemma 59 (Two-Derivative Matching) *On the image of F the heterotic non-linear sigma-model metric restricted to the visible E_6 -breaking directions coincides, up to an overall normalization f^2 , with the pull-back $F^*\mathfrak{g}$ of the unique E_6 -invariant EIII metric $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$.*

Proof The residual visible directions transform in $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ under H , which is real-irreducible (Lemma 21). Any H -invariant symmetric two-tensor on this representation is unique up to scale by Schur’s lemma; both the sigma-model metric and $\mathfrak{g}$ are such tensors, hence proportional. The proportionality constant defines f^2 . $\square$

Lemma 60 (Current-Algebra Matching) *The $H = \text{Spin}(10) \times \text{U}(1)_\psi$ isometry action on $\mathcal{M}$ is generated by the same currents as the level-one $\text{Spin}(10)$ and $\text{U}(1)_\psi$ Kač–Moody currents of the heterotic worldsheet theory, with matched central data. In particular the $\text{U}(1)_\psi$ charge assignment $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ reproduces the worldsheet charge lattice of the visible matter.*

Table 2 Correspondence dictionary between the visible heterotic sector and the coset sector. The correspondence is established at the level of the two-derivative bosonic action and the $\text{Spin}(10) \times \text{U}(1)_\psi$ current algebra.

Heterotic $E_8 \times E_8$ visible sector	coset sector
E_6 current algebra (level one)	Compact E_6 symmetry selected by A
Adjoint Higgs orbit $E_6 \cdot \langle \mathbf{78}_H \rangle$	Vacuum manifold $\mathcal{M} = E_6 / (\text{Spin}(10) \times \text{U}(1))$
Visible matter $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$	Goldstone tangent space $T\mathcal{M}$
Sigma-model metric on visible moduli	EIII metric $\mathfrak{g} = f^2(-K)$
$\text{Spin}(10) \times \text{U}(1)_\psi$ Kač–Moody currents	Isometry generators of $\mathcal{M}$
Green–Schwarz axion	$\text{U}(1)_\psi$ Goldstone mode
No fundamental visible graviton	Induced (Sakharov–Adler) gravity

Proof The Killing vector fields of $\mathcal{M}$ generating H act on the tangent representation exactly as the zero modes of the $\text{Spin}(10) \times \text{U}(1)_\psi$ currents act on the visible states; the charges agree by construction of F , and the $\text{U}(1)_\psi$ normalization is fixed by requiring the $\mathbf{16}$ to carry charge -3 , matching the branching $\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$. $\square$

Theorem 61 (Conditional Local Heterotic Realization) *Assume the heterotic correspondence map F of Definition 58 exists for a heterotic background satisfying the stated current-algebra data. Then, in a neighbourhood of the gauge-unfixed Higgs orbit, the two-derivative bosonic action of the visible E_6 -breaking sector of the heterotic model is mapped by F onto the coset action of § 3, with the induced Einstein–Hilbert term of § 4 providing the gravitational sector. Under this map the Green–Schwarz axion [27] of the heterotic background is identified with the $\text{U}(1)_\psi$ Goldstone mode of $\mathcal{M}$.*

Proof Combine Lemma 59 and Lemma 60: the kinetic term matches by two-derivative matching and the symmetry action matches by current-algebra matching, so the visible bosonic Lagrangians agree to two derivatives. The gravitational sector is not present in the heterotic two-derivative visible action as an independent field; on the coset side it is induced (§ 4), consistent with the absence of a fundamental visible-sector graviton. The Green–Schwarz axion couples to $\text{Tr } F \wedge F$ with a shift symmetry, and the correspondence identifies the corresponding visible shift mode with the $\text{U}(1)_\psi$ Goldstone of $\mathcal{M}$. $\square$

Remark 62 (Scope of the Correspondence) Theorem 61 is a statement about the two-derivative bosonic action and the current algebra only. It does not assert an exact conformal duality, a matching of the full string spectrum, or agreement of higher-derivative α' corrections. In particular the heterotic side carries an infinite tower of massive string states with no coset counterpart, and the coset side is an effective field theory valid below the string scale. The value of the correspondence is structural: it exhibits the coset as the low-energy image of a genuine string background, and it shows that the induced-gravity mechanism is compatible with a formulation in which no fundamental visible-sector graviton is postulated.

We now consider a heterotic compactification intended to meet the criteria of Theorem 61: a $\mathbb{Z}_{12}$ asymmetric orbifold of the $E_8 \times E_8$ heterotic string [28, 29] whose

visible sector is a three-generation E_6 grand unified theory with an adjoint Higgs $\mathbf{78}_H$ [30]. Giving the adjoint Higgs a vacuum expectation value in the $E_6 \supset H$ singlet direction breaks $E_6 \rightarrow H = \text{Spin}(10) \times U(1)_\psi$, and the local orbit computation identifies the resulting gauge-unfixed Higgs orbit with the coset $\mathcal{M}$.

The model is a $\mathbb{Z}_{12}$ asymmetric orbifold with visible gauge group E_6 realized at level one, three chiral generations in the $\mathbf{27}$, and an adjoint Higgs multiplet $\mathbf{78}_H$ arising from the diagonal embedding of the visible E_6 in the ten-dimensional gauge group. Under $E_6 \supset H = \text{Spin}(10) \times U(1)_\psi$ the adjoint branches as

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (13)$$

and we take the Higgs vacuum $\langle \mathbf{78}_H \rangle = \Phi_0$ to lie in the $\mathbf{1}_0$ direction.

Theorem 63 (Conditional Gauge-Unfixed EIII Orbit Theorem) *Assume the $\mathbb{Z}_{12}$ model contains an adjoint-Higgs vacuum Φ_0 in the $\mathbf{1}_0 \subset \mathbf{78}$ direction under $E_6 \supset H = \text{Spin}(10) \times U(1)_\psi$. Then the gauge-unfixed Higgs orbit is*

$$\mathcal{O}_H = E_6 \cdot \Phi_0 \cong E_6 / (\text{Spin}(10) \times U(1)) = \mathcal{M}, \quad T_{\mathbb{C}} \mathcal{O}_H = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}.$$

Proof By the branching (13) the centralizer in E_6 of a generic vector in the $\mathbf{1}_0$ direction is exactly $H = \text{Spin}(10) \times U(1)_\psi$: the $\mathbf{45}_0$ commutes with Φ_0 , the $U(1)_\psi$ generator is Φ_0 itself, and the $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ carry non-zero $U(1)_\psi$ charge and are therefore rotated. By the orbit-stabilizer theorem $E_6 \cdot \Phi_0 \cong E_6 / H = E_6 / (\text{Spin}(10) \times U(1))$. The tangent space at Φ_0 is the span of the broken generators, namely $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, giving the stated complexified tangent representation. This is the coset $\mathcal{M}$ of § 3. $\square$

Theorem 64 (Gauge-Fixed/Gauge-Unfixed Equivalence) *The EIII scalar sector is physically equivalent to the gauge-unfixed Stueckelberg/Higgs-orbit description of the $E_6 \rightarrow \text{Spin}(10) \times U(1)_\psi$ breaking sector of the $\mathbb{Z}_{12}$ model. Unitary gauge maps this description to the standard massive-vector description of the heterotic E_6 grand unified theory, in which the $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ Goldstones are absorbed by the broken gauge bosons.*

Proof In the gauge-unfixed description the 32 real coordinates of $\mathcal{O}_H \cong \mathcal{M}$ are Stueckelberg/Goldstone fields for the broken E_6/H directions. Fixing unitary gauge sets the Goldstones to zero and gives mass to the corresponding gauge bosons via the Higgs mechanism; the massive vector content is gauge-invariant and hence the same in both descriptions. The coset action of § 3 is the gauge-unfixed Goldstone dynamics on $\mathcal{O}_H$, so the two descriptions are equivalent at the level of this sector. $\square$

Corollary 65 (Breaking Chain) *The $\mathbb{Z}_{12}$ model realizes the phenomenological breaking chain*

$$E_6 \rightarrow \text{Spin}(10) \times U(1)_\psi \rightarrow \text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times U(1),$$

with the first step realized geometrically as the emergence of the coset $\mathcal{M}$ and the subsequent steps by the standard heterotic Higgs mechanism.

Proof The first arrow is [Theorem 63](#). The remaining arrows are the standard $\text{Spin}(10) \rightarrow \text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ descent of a heterotic E_6 grand unified theory with the matter content of the [27](#); three generations follow from the three fixed-point families of the $\mathbb{Z}_{12}$ action [\[30\]](#). $\square$

Corollary 66 (Conditional UV Completion) *If the $\mathbb{Z}_{12}$ model of [\[30\]](#) satisfies the global heterotic consistency conditions listed in [Remark 68](#), then it is an ultraviolet completion of the effective theory in the sense of the local correspondence of [Theorem 61](#).*

Proof Immediate from [Theorem 63](#), [Theorem 61](#), and the definition of the global heterotic consistency conditions: local matching plus global worldsheet consistency is the required condition for the heterotic model to define a consistent string background whose low-energy visible sector is the coset theory. $\square$

The results above give a relative swampland criterion. We do not claim that all non-coset vacua are inconsistent. The statement is restricted to the Albert-selected heterotic class: within that class, a vacuum can realize the coset correspondence only when its gauge-unfixed Higgs/moduli sector is EIII with the data specified below.

Theorem 67 (Albert–EIII Swampland Criterion) *Let $\mathcal{T}$ be a four-dimensional effective theory proposed as the Albert-selected low-energy sector of an $E_8 \times E_8$ heterotic compactification. Assume that its visible exceptional sector is selected by the Albert algebra $\mathbb{A}$ and that the corresponding compact real form is E_6 . Then $\mathcal{T}$ lies in the Albert-selected heterotic landscape only if its gauge-unfixed Higgs/moduli sector contains a residual scalar orbit*

$$\mathcal{O} \cong E_6/(\text{Spin}(10) \times \text{U}(1)) = \mathcal{M}, \quad T_{\mathbb{C}} \mathcal{O} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3},$$

carrying the unique positive E_6 -invariant EIII metric, globally consistent $\text{Spin}(10) \times \text{U}(1)_{\psi}$ gauge data, anomaly cancellation, spin-lift compatibility, Green–Schwarz matching, and heterotic worldsheet consistency. Equivalently, any proposed Albert-selected four-dimensional effective theory failing one of these conditions lies outside the Albert–EIII landscape: it has no $E_8 \times E_8$ heterotic ultraviolet completion of the specified type realizing the coset correspondence.

Proof The argument is by necessity together with the contrapositive.

Albert selection fixes the complex exceptional type to E_6 ([§ 2](#)); choosing the compact real form gives the compact E_6 structure used in the EIII target and guarantees a positive-definite invariant metric ([Theorem 22](#)).

Imposing the grand-unified isotropy $H = \text{Spin}(10) \times \text{U}(1)_{\psi}$ forces the homogeneous scalar orbit to be E_6/H . By the orbit–stabilizer theorem an adjoint Higgs vacuum in the $\mathbf{1}_0$ direction of the branching [\(13\)](#) has centralizer H , so its gauge-unfixed Higgs orbit is exactly $E_6 \cdot \Phi_0 \cong E_6/(\text{Spin}(10) \times \text{U}(1))$ ([Theorem 63](#)).

The tangent representation is forced to be $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, which is real-irreducible. Hence the H -invariant positive metric is unique up to scale, and there is no competing compact homogeneous residual scalar sector with the same Albert/ E_6/H data ([Lemma 21](#), [Theorem 22](#)).

Because $\mathfrak{m}$ has no invariant four-plane and the EIII metric is positive definite, the coset cannot supply a Lorentzian vierbein by selecting four invariant scalar directions.

The valid construction is the nondegenerate Euclidean pull-back of [Theorem 17](#), followed by Osterwalder–Schrader reconstruction; gravity then enters through the induced Sakharov–Adler dynamics of [§ 4](#).

Local EIII matching is not sufficient; a heterotic completion must further satisfy modular invariance, GSO/fermion-parity consistency, flux and Wilson-line quantization, bundle transition-function compatibility, anomaly cancellation on overlaps, spin-lift conditions for the **16** matter, and Green–Schwarz matching [\[27\]](#).

If a proposed effective theory lacks the EIII orbit, the required tangent representation, the EIII metric, the correct gauge action, the anomaly/Green–Schwarz compatibility, the spin-lift data, or global heterotic consistency, then it cannot be the Albert-selected $E_8 \times E_8$ ultraviolet completion of the specified type. $\square$

Remark 68 (Global Heterotic Consistency Ledger) The global heterotic consistency conditions invoked throughout [§ 7](#) are: modular invariance, GSO/fermion-parity consistency, flux and Wilson-line quantization, bundle transition-function compatibility, anomaly cancellation on overlaps, spin-lift conditions for the 16 matter, and Green–Schwarz matching [\[27\]](#).

Corollary 69 ($\mathbb{Z}_{12}$ Test) *A $\mathbb{Z}_{12}$ asymmetric heterotic E_6 orbifold realizes the coset correspondence if its gauge-unfixed adjoint-Higgs orbit is $E_6/(\text{Spin}(10) \times \text{U}(1))$ and if its full worldsheet, anomaly, bundle, flux, spectrum, renormalization-group, and gravitational matching data satisfy the global heterotic consistency conditions of [Remark 68](#). If these checks fail, the model remains an E_6 -aligned heterotic model but not the coset ultraviolet completion.*

Proof The orbit condition is [Theorem 63](#) and places the model in the Albert-selected landscape of [Theorem 67](#); the global conditions are those of [Remark 68](#) and [Corollary 66](#). If any global condition fails, the contrapositive of [Theorem 67](#) applies. $\square$

Corollary 70 ($\mathbb{Z}_{12}$ Conditional Candidate Criterion) *The $\mathbb{Z}_{12}$ asymmetric E_6 orbifold is a concrete candidate for realizing the coset correspondence because it supplies an $E_8 \times E_8$ heterotic origin, a visible E_6 grand unified sector, three generations, and an adjoint Higgs obtained by diagonal embedding. If its gauge-unfixed adjoint-Higgs orbit, anomaly data, worldsheet data, and gravitational reduction satisfy the global heterotic equivalence criterion of [Remark 68](#), then it gives a heterotic ultraviolet completion of the coset theory.*

Proof The four structural features are those of the model of [\[30\]](#) recorded in [Theorem 63](#); each is a prerequisite of [Theorem 67](#), and no other presently identified construction is known to supply all four together with the EIII orbit of [Theorem 63](#). $\square$

Remark 71 (Status of the Criterion) [Theorem 67](#) is a relative, class-restricted statement: it constrains only theories proposed as Albert-selected heterotic vacua. It is not a universal assertion that all other four-dimensional effective theories are inconsistent. Within its stated scope it is sharp, and it makes the $\mathbb{Z}_{12}$ model of [\[30\]](#) a concrete landscape candidate subject to the checklist of [Corollary 69](#).

8 Discussion

The passage from the coset sector to observable physics is constrained first by anomaly cancellation. This is already a point at which the coset truncation is not, by itself, a complete four-dimensional theory. The low-energy field content in [Table 1](#) gauges $H = \text{Spin}(10) \times \text{U}(1)_\psi$ and contains chiral matter in a single $\mathbf{16}_{+1}$. Although the $\text{Spin}(10)$ factor is anomaly-free in four dimensions, the abelian factor is not. A lone $\mathbf{16}_{+1}$ leaves mixed and abelian anomalies of types

$$[\text{Spin}(10)]^2 \text{U}(1)_\psi, \quad [\text{U}(1)_\psi]^3, \quad \text{grav}^2 \text{U}(1)_\psi.$$

These anomalies cancel identically for a complete

$$\mathbf{27} = \mathbf{16}_{+1} \oplus \mathbf{10}_{-2} \oplus \mathbf{1}_{+4}.$$

Thus the phenomenological interpretation of the truncation must be made through the heterotic boundary, where anomaly cancellation is supplied by one of two mechanisms.

The first possibility is the four-dimensional Green–Schwarz mechanism [\[27, 31\]](#). By [Theorem 61](#), the $\text{U}(1)_\psi$ Goldstone mode of $\mathcal{M}$ is identified with the Green–Schwarz axion of the heterotic background. Its induced couplings to $\text{Tr}(F \wedge F)$ shift under $\text{U}(1)_\psi$ and cancel the residual anomalies listed above. The same mechanism gives a Stückelberg mass, of order gf , to the $\text{U}(1)_\psi$ gauge boson. Below this scale the effective gauge group is the anomaly-free $\text{Spin}(10)$, and the breaking chain of [Corollary 65](#) proceeds in the usual grand-unified manner.

The second possibility is spectrum completion. In the $\mathbb{Z}_{12}$ realization of [\[30\]](#), the chiral matter appears in three complete $\mathbf{27}$ generations. The four-dimensional chiral spectrum is then anomaly-free without relying on the anomalous $\text{U}(1)_\psi$ cancellation alone, while the Green–Schwarz mechanism remains part of the universal heterotic anomaly descent. In this sense, the coset truncation fixes the representation-theoretic seed of one family, whereas family replication and complete anomaly cancellation are properties of the ultraviolet completion.

The resulting phenomenology is therefore that of an $\text{Spin}(10)$ grand unified theory subject to heterotic matching. The matching scale obtained in [§ 5](#) is

$$M_{\text{GUT}} \simeq 2 \times 10^{16} \text{ GeV},$$

with the qualification that this scale is a consistency output of the heterotic handoff, not a prediction of the nonsupersymmetric coset flow alone. Dimension-six gauge-boson exchange then gives proton-decay channels with partial lifetimes in the range

$$10^{35}\text{--}10^{37} \text{ yr},$$

with the lower part of this interval within the projected reach of next-generation searches [\[32\]](#). The precise lifetime depends on the threshold spectrum at the matching scale, the normalization of the unified coupling, and the hadronic matrix elements

entering the decay amplitudes. The construction should therefore be read as restricting a proton-decay window, rather than predicting a single lifetime.

Neutrino physics provides a second standard phenomenological channel. Each **16** contains a right-handed neutrino, so a seesaw mechanism becomes available after $\text{Spin}(10)$ breaking. The Majorana scale is not fixed by the Albert–EIII kinematics; it depends on the ultraviolet completion and on the operators responsible for the descent from $\text{Spin}(10)$ to the Standard Model gauge group. Nevertheless, the embedding constrains the allowed charge assignments and places the neutrino sector in the same unified multiplet structure as the charged matter fields.

The heavy states associated with the first breaking step are not expected to be directly accessible at low energy. The Z'_ψ associated with the Stückelberg-massive $U(1)_\psi$ lies near the matching scale. Likewise, the 28 degenerate transverse coset scalars of [Theorem 26](#), located at the top of the hierarchy described in [§ 4](#), are structural features of the ultraviolet matching rather than ordinary TeV-scale scalar states. Their significance is instead diagnostic: the degeneracy follows from the H -irreducibility of the tangent representation and would be lifted only by effects beyond the invariant one-loop truncation.

The gravitational sector has a similarly restricted interpretation. The metric induced in [Theorem 17](#) is Riemannian, and Lorentzian physics is recovered only after Osterwalder–Schrader reconstruction, as described in [Remark 18](#). The Einstein–Hilbert term is induced at one loop in the Sakharov–Adler sense ([Theorem 25](#)), while higher-curvature operators are irrelevant only inside the effective window of [Theorem 27](#). The same heat-kernel expansion also produces an a_0 vacuum-energy term. In the present construction this term is handled by matching; no dynamical relaxation mechanism is supplied. Thus the cosmological-constant problem is not solved by the coset construction. It is transferred to the boundary data relating the effective coset theory to its heterotic completion.

The zeta sector is even more conditional. The results of [§ 6](#) give a spectral criterion, not a construction of the required spectral realization. Theorems such as [Theorem 46](#) show that a positive Galois-equivariant realization of the shifted zeta divisor inside the coset spectral subalgebra would imply the Riemann hypothesis. They do not prove that such a realization exists. The continuous bridge of [Theorem 57](#) is therefore best understood as a precise formulation of the obstruction: one must construct a compatible family of positive self-adjoint mass-squared operators whose arithmetic endpoint has the prescribed shifted zeta divisor.

These observations also clarify the status of the universality claim. Once the Albert–EIII datum is fixed, the vacuum orbit, tangent representation, invariant metric, induced Riemannian geometry, and one-loop degeneracy pattern are fixed. The data not fixed by the coset are exactly the data required at the boundaries of the effective theory: reflection positivity for the constructive operator-algebraic formulation, heterotic consistency for the ultraviolet completion, and a positive arithmetic spectral realization for the zeta criterion. The theory is universal in its structural sector, not in the sense of supplying these boundary data automatically.

Several problems remain open.

First, the operator-algebraic construction should be made constructive. The Haag–Kastler net in § 2 is characterized through its Schwinger functions, but reflection positivity of the interacting four-dimensional nonlinear σ -model is assumed. A natural program is to formulate the EIII model on a lattice with exact global E_6 invariance and to analyze reflection positivity, clustering, and the continuum limit directly. The compactness of the target and the uniqueness of the invariant metric make the model rigid enough for such an analysis, although no constructive result is presently available.

Second, the signature issue should be separated from the Euclidean construction. The present paper uses the positive EIII metric and obtains Lorentzian physics only through reconstruction. A genuinely Lorentzian formulation would require an explicit signature-selecting term, for example through the coset-sector order parameter of ???. The corresponding problem is to classify the H -invariant signature-selecting operators and determine whether the breaking

$$\mathrm{SO}(4) \longrightarrow \mathrm{SO}(1,3)$$

can be made dynamically preferred.

Third, the heterotic ledger must be completed for the candidate $\mathbb{Z}_{12}$ model. The local orbit statement of Theorem 63 and the local correspondence theorem Theorem 61 are not sufficient by themselves. One must verify modular invariance, GSO consistency, flux and Wilson-line quantization, bundle transition functions, spin lifts for the **16** matter, Green–Schwarz matching, and the spectrum match to three **27** generations and the adjoint Higgs. This is precisely the checklist isolated in Remark 68. Until those checks are completed, Corollary 66 remains conditional on the global consistency of the candidate background.

Fourth, the matching calculation should be sharpened. The one-loop estimate of § 5 should be extended to two loops with explicit thresholds at the $\epsilon = 1/6$ boundary. This would refine both the allowed band for M_{GUT} and the associated proton-lifetime window. It would also test whether the threshold spectrum required by the heterotic handoff is compatible with the scalar degeneracy and induced Planck-scale hierarchy of the coset sector.

Fifth, finite-level zeta realizations should be studied before the full adelic problem. One may attempt to construct candidate operators $\mathcal{M}_b^2$ using finitely many primes and truncated divisors, verify Galois compatibility at each finite level, and then examine whether norm-resolvent control survives the limiting procedure. A finite-level obstruction would be as informative as a successful approximation, since it would identify precisely where positivity, Galois compatibility, or spectral continuity fails.

Finally, the cosmology of the coset sector remains largely undeveloped. The 28 degenerate transverse scalars form a constrained multi-field sector with a unique invariant metric and a common one-loop mass. This severely restricts the admissible inflationary dynamics, but does not by itself determine the vacuum energy. The natural cosmological problem is therefore not merely to write an inflationary potential, but to understand how the induced vacuum-energy term is matched across the heterotic boundary and whether that matching can be made stable under the same global consistency conditions that control anomaly cancellation.

The preceding points summarize the role of the present construction. The Albert–EIII datum fixes a rigid coset kinematics and an induced geometric sector. It does not, by itself, construct the interacting Euclidean theory, select Lorentzian signature dynamically, solve the cosmological-constant problem, prove the existence of the zeta realization, or verify the global heterotic ledger. Those are the remaining boundary problems. Closing them would turn the conditional rigidity results proved here into a complete four-dimensional model.

9 Conclusion

We have formulated UCFT as a four-dimensional operator-algebraic quantum field theory determined, conditionally but rigidly, by a single exceptional datum: the Albert algebra with its compact E_6 symmetry. Two postulates select the EIII coset sector, and from that point the kinematics admit no further choices. The vacuum orbit is $E_6/(\text{Spin}(10) \times U(1))$; its 32 Goldstone directions form the irreducible $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ with a unique positive invariant metric; spacetime geometry arises as the first fundamental form of a spacetime-filling configuration; and the Einstein–Hilbert term is induced at one loop rather than postulated.

Appendix A Albert Algebra Proofs

This appendix supplies proofs regarding the Albert algebra deferred from § 2.

Proof of Theorem 5 Choose a basis $\{e_0, e_1, \dots, e_7\}$ of the octonions satisfying $e_0 = 1$ and $e_i^2 = -1$ for $i \geq 1$. Every $X \in \mathbb{A}$ can be written uniquely as the Hermitian 3×3 matrix

$$X = \begin{pmatrix} \alpha_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \alpha_2 & x_1 \\ x_2 & \bar{x}_1 & \alpha_3 \end{pmatrix}, \quad \alpha_i \in \mathbb{R}, \quad x_i \in \mathbb{O}.$$

The real-dimension count is therefore $3 \text{ scalars} + 3 \times 8 \text{ octonion components} = 27$. Jordan commutativity is obvious from the symmetric product $\circ$. Because $\mathbb{O}$ is nonassociative, $\mathbb{A}$ is *not* a subalgebra of an associative algebra under $X \circ Y = \frac{1}{2}(XY + YX)$, so the Jordan identity $a \circ (b \circ a^2) = (a \circ b) \circ a^2$ is not automatic. It nevertheless holds by the Jordan–von Neumann–Wigner classification of Hermitian matrix algebras over composition algebras: for $n \leq 3$ the octonionic alternative law suffices to verify the degree-four identity on $\mathfrak{h}_n(\mathbb{O})$, whereas for $n \geq 4$ it fails [1, 3, 33]. Hence $\mathbb{A} = \mathfrak{h}_3(\mathbb{O})$ is a 27-dimensional formally real Jordan algebra; being nonspecial, it is the exceptional one. $\square$

Proof of Theorem 6 For X as above, set

$$Q(X) = \text{Tr}(X^2), \quad N(X) = \alpha_1 \alpha_2 \alpha_3 - \sum_{i=1}^3 \alpha_i \|x_i\|^2 + 2 \Re(x_1 x_2 x_3),$$

where juxtaposition is octonion multiplication. The form Q is quadratic and N is cubic in the real coordinates. Direct calculation shows $N(g \cdot X) = \lambda(g)N(X)$ for every $g \in \text{Str}(\mathbb{A})$ with multiplicative character $\lambda(g) \in \mathbb{R}^\times$ [1]; the norm-one subgroup $\{\lambda(g) = 1\}$ is the noncompact real form $E_{6(-26)}$ and preserves N . The trace form Q is preserved by the automorphism group $F_4 = \text{Aut}(\mathbb{A})$, which fixes the unit element and the Jordan product; it is not preserved by

the full reduced structure group, since the **27** of E_6 admits no invariant bilinear form (**27** is not self-dual). On the complexification $J_3(\mathbb{O})_{\mathbb{C}}$, the compact real form E_6 is by definition the subgroup of the complex norm-preserving group that is unitary with respect to the Hermitian form $\langle X, Y \rangle = \text{Tr}(X \circ \bar{Y})$; it therefore preserves N and $\langle \cdot, \cdot \rangle$ simultaneously [2]. $\square$

Proof of Theorem 7 Let $\text{Str}^0(\mathbb{A})$ denote the connected component of the identity in $\text{Str}(\mathbb{A})$. Jacobson's construction [1] realizes the exceptional structure algebra as the sum of derivations of $\mathbb{A}$ (an $\mathfrak{f}_4$) plus the traceless right multiplications

$$R_X : Y \mapsto X \circ Y - \frac{1}{3}Q(X, Y)\mathbf{1}.$$

After complexification this Lie algebra has type $\mathfrak{e}_6$. The main text uses the simply connected compact real form of the same complex Lie type for the EIII homogeneous space; this is the group denoted E_6 throughout the coset construction. $\square$

Lemma 72 *If $g \in E_6$ and $X \in J_3(\mathbb{O})_{\mathbb{C}}$ then $\langle gX, gX \rangle = \langle X, X \rangle$ and $\langle (gX)^{\sharp}, (gX)^{\sharp} \rangle = \langle X^{\sharp}, X^{\sharp} \rangle$.*

Proof Invariance of the Hermitian form is Theorem 6. The adjugate satisfies the structure-group equivariance $(gX)^{\sharp} = g^{\sharp}(X^{\sharp})$, where $g \mapsto g^{\sharp}$ is the dual (contragredient) representation on **27** [2, 3]; the compact form acts unitarily on the dual as well, so the Hermitian norm of $X^{\sharp}$ is invariant. $\square$

Proof of Theorem 8 The locus $\{X^{\sharp} = 0\} \setminus \{0\}$ is the cone of rank-one elements of $J_3(\mathbb{O})_{\mathbb{C}}$, i.e. the affine cone over the unique closed orbit of the complex group of type E_6 in $\mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}})$ (the complexified Cayley plane) [2]. For a linear action of a compact group whose complexification acts on $\mathbb{P}(V)$, the closed complex orbit is a single orbit of the compact real form: it is the orbit of the highest-weight line [5]. Hence E_6 acts transitively on rank-one lines, and by Corollary 9 the stabilizer of a line is the maximal-rank subgroup $H = \text{Spin}(10) \times \text{U}(1)$; the orbit of lines is $\text{EIII} = E_6/H$.

Under H the **27** branches as $\mathbf{16}_{+1} \oplus \mathbf{10}_{-2} \oplus \mathbf{1}_{+4}$, and the highest-weight line is the $\text{Spin}(10)$ -singlet direction $\mathbf{1}_{+4}$, which carries nonzero $\text{U}(1)_{\psi}$ charge. Therefore $\text{Spin}(10)$ fixes a highest-weight vector X_0 pointwise, while the $\text{U}(1)_{\psi}$ factor acts on the line $\mathbb{C}X_0$ through a nontrivial character and hence acts transitively on the circle of vectors of fixed Hermitian norm in that line. Combining the two transitivityes, E_6 acts transitively on S_v , with vector stabilizer isomorphic to $\text{Spin}(10)$; thus $S_v \cong E_6/\text{Spin}(10)$, of dimension $78 - 45 = 33$, a circle bundle over EIII. $\square$

Lemma 73 *For $\alpha > 0$, $k \geq 2$ even, the polynomial $V(X) = P(X) + \alpha[\langle X, X \rangle - c_2]^k$ satisfies $V(X) \rightarrow \infty$ as $\|X\| \rightarrow \infty$.*

Proof The Hermitian form is exactly the squared norm, $\langle X, X \rangle = \|X\|^2$. Since k is even,

$$\alpha[\langle X, X \rangle - c_2]^k = \alpha\|X\|^{2k} + O(\|X\|^{2k-2}),$$

so V is radially unbounded. $\square$

Lemma 74 $P(X) \geq 0$ for all $X \in J_3(\mathbb{O})_{\mathbb{C}}$, with equality if and only if $X \in S_v = E_6 \cdot X_0$.

Proof Non-negativity is clear from (1), both summands being squared norms and squares. Equality forces

$$X^\sharp = 0 \quad \text{and} \quad \langle X, X \rangle = c_2;$$

Theorem 8 then yields the stated orbit condition. $\square$

Appendix B Emergent Spacetime Proofs

This appendix supplies proofs for Lemma 16 and Theorem 17 from § 3.

Recall that the broken directions of the coset

$$\mathcal{M} = \frac{E_6}{\text{Spin}(10) \times \text{U}(1)}$$

form a 32-dimensional real vector space $\mathfrak{m}$ carrying the unique invariant metric $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$ of Theorem 22, which on the compact real form of E_6 is positive definite. Let

$$d\phi_x : T_x \mathbb{R}^4 \rightarrow T_{\phi(x)} \mathcal{M} \cong \mathfrak{m}$$

denote the differential of the map $\phi^\alpha(x)$.

Proof of Lemma 16 The pull-back at x is the quadratic form

$$g_{\mu\nu}(x) = \mathfrak{g}_{\alpha\beta} \partial_\mu \phi^\alpha \partial_\nu \phi^\beta = \mathfrak{g}(d\phi_x(\partial_\mu), d\phi_x(\partial_\nu)).$$

Because $\mathfrak{g}$ is positive definite, the associated Gram map obeys $\text{rank } g_{\mu\nu} = \text{rank } d\phi_x = \dim V_x$, where $V_x = \Im d\phi_x \subseteq \mathfrak{m}$: indeed $g_{\mu\nu} v^\mu = 0$ iff $\mathfrak{g}(d\phi_x v, \cdot) = 0$ iff $d\phi_x v = 0$ by definiteness. Since $\dim T_x \mathbb{R}^4 = 4$ we have $\dim V_x \leq 4$, and non-degeneracy of $g_{\mu\nu}$ forces $\dim V_x = 4$, i.e. $d\phi_x$ is injective and $V_x = \mathfrak{p}_0(x)$ is a four-plane. Its $\mathfrak{g}$ -orthogonal complement $\mathfrak{p}_m(x)$ has dimension 28. $\square$

Proof of Theorem 17 Embed $\text{Spin}(10) \times \text{U}(1)$ in E_6 as the stabilizer of the rank-one highest-weight line $[X_0] \in \mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}})$ (Corollary 9, Theorem 8), and write $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$. On the compact real form of E_6 the Killing form is negative definite on all of $\mathfrak{e}_6$; hence $-K$, and therefore $\mathfrak{g} = f^2(-K)$, is positive definite on $\mathfrak{m}$. (This is the same fact that makes the symmetric space $\mathcal{M} = \frac{E_6}{\text{Spin}(10) \times \text{U}(1)}$ Riemannian; EIII is Hermitian symmetric, hence Riemannian, and carries no invariant indefinite metric.)

By Lemma 16 the tetrad $e_\mu^a = \theta_\alpha^a \partial_\mu \phi^\alpha$ is invertible for a non-degenerate configuration. Choosing a $\mathfrak{g}$ -orthonormal basis $\{u_a\}_{a=0}^3$ of $\mathfrak{p}_0(x)$, so that $\mathfrak{g}(u_a, u_b) = \delta_{ab}$, and decomposing $\partial_\mu \phi^\alpha = e_\mu^a u_a^\alpha$,

$$g_{\mu\nu} = \mathfrak{g}(u_a, u_b) e_\mu^a e_\nu^b = \delta_{ab} e_\mu^a e_\nu^b,$$

which is symmetric, non-degenerate (invertibility of e_μ^a), and positive definite. Smoothness in x follows from smoothness of ϕ and of the Maurer-Cartan form. Openness and density of the non-degenerate locus hold because $\det e_\mu^a \neq 0$ is an open condition satisfied by generic immersions. Thus $g_{\mu\nu}$ is a Riemannian metric on $\mathbb{R}^4$, namely the first fundamental form of ϕ .

The signature is (4, 0), not (1, 3): a pull-back of a positive-definite form is positive semidefinite for every ϕ , so no choice of configuration can produce a Lorentzian induced metric. $\square$

Appendix C Heat-Kernel Coefficients and Induced Gravity

This appendix proves [Theorem 22](#), [Theorem 25](#), and [Theorem 26](#) from § 4: the uniqueness of the invariant coset metric, the one-loop induced Einstein–Hilbert term, and the mass spectrum of the coset scalars.

Proof of Theorem 22 Write the Cartan decomposition $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$. For $g \in E_6$ define the adjoint action $\text{Ad}_g(X) = gXg^{-1}$. The isotropy module $\mathfrak{m}$ is real-irreducible of complex type ([Lemma 21](#)): its commutant is $\mathbb{C} = \text{span}\{\mathbf{1}, J\}$, with J the invariant complex structure. An invariant symmetric bilinear form on $\mathfrak{m}$ has the shape $b(X, Y) = -K(TX, Y)$ with T in the commutant, $T = a\mathbf{1} + bJ$. Since J is K -antisymmetric, the J -part of b is the (antisymmetric) Kähler form; symmetry of b forces $b = 0$ there, so b is proportional to the restriction of the Killing form, $\mathfrak{g}_{\alpha\beta} = cK_{\alpha\beta}$. On the compact real form of E_6 the Killing form is negative definite, so $K|_{\mathfrak{m}} < 0$; positivity of the σ -model kinetic term therefore fixes $c = -f^2 < 0$, i.e. $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta}) > 0$, completing the proof. $\square$

Gauge fixing for the local $H = \text{Spin}(10) \times \text{U}(1)$ symmetry follows the Faddeev–Popov procedure [[15](#)]. The collective matter multiplet is $\Phi = (\phi^\alpha, A_\mu^i, \psi)$, with $\alpha \in \{1, \dots, 32\}$ and $i \in \{1, \dots, 46\}$. Here, ϕ^α are the real coset-scalar fluctuations, A_μ^i the gauge-field fluctuations, and ψ the Weyl fermions. Ghost, antighost, and Nakanishi–Lautrup fields are denoted by $(c^i, \bar{c}^i, B^i)$. With the background Lorenz gauge $\mathcal{F}_i = \nabla^\mu A_{\mu i}$, the gauge-fixed action reads

$$S_{\text{BRST}} = S[\Phi] + \int d^4x \sqrt{-g} \left[B^i \mathcal{F}_i + \frac{\xi}{2} B^i B_i + \bar{c}^i \Delta_{\text{ghost}} c_i \right], \quad (\text{C1})$$

$$\Delta_{\text{ghost}} = -\nabla^2 \delta_{ij} + R_{ij}, \quad R_{ij} = f_{ij}{}^k F_{\mu\nu k} \gamma^\mu \gamma^\nu.$$

The nilpotent BRST operator s acts as

$$\begin{aligned} s\phi^\alpha &= -c^i (t_i)^\alpha_\beta \phi^\beta, & sA_\mu^i &= D_\mu c^i, \\ D_\mu c^i &= \partial_\mu c^i + f_{jk}^i A_\mu^j c^k, & s\psi &= i c^i T_i \psi, \\ s c^i &= -\frac{1}{2} f_{jk}^i c^j c^k, & s\bar{c}^i &= B^i, \\ s B^i &= 0, \end{aligned}$$

so $s^2 = 0$ and physical correlators are gauge-independent. Here, B^i is the Nakanishi–Lautrup auxiliary field, and $\gamma^\mu = e_a^\mu \gamma^a$ are the curved-space Dirac matrices obeying $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$. Eliminating B^i by its algebraic equation of motion reproduces the usual gauge-fixing term $(2\xi)^{-1}(\mathcal{F}_i)^2$ and leaves the ghost operator Δ_{ghost} defined above.

Proof of Theorem 25 Let $\bar{\Phi}(x)$ be a fixed classical background and define the full fluctuation multiplet

$$\Phi(x) = (\phi^\alpha(x), A_\mu^i(x), \psi^a(x), c^i(x), \bar{c}^i(x)),$$

with $\alpha \in \{1, \dots, 32\}$ and $i \in \{1, \dots, 46\}$. The ghosts $c^i, \bar{c}^i$ transform in the adjoint of H . Writing $\Phi = \bar{\Phi} + \varphi$, the gauge-fixed BRST action (C1) expands as

$$S_{\text{BRST}}[\bar{\Phi} + \varphi] = S_{\text{BRST}}[\bar{\Phi}] + S^{(2)}[\varphi] + O(\varphi^3),$$

with quadratic part

$$S^{(2)}[\varphi] = \frac{1}{2} \int d^4x \sqrt{-g} \varphi^T \Delta \varphi, \quad \Delta = \begin{pmatrix} -\nabla^2 \mathbf{1}_{32} + M_\phi^2 & 0 & 0 \\ 0 & -\nabla^2 \mathbf{1}_{46} + M_A^2 & 0 \\ 0 & 0 & i\nabla\!\!\!/ \end{pmatrix}, \quad (\text{C2})$$

where $(M_\phi^2)_{\alpha\beta} = [\text{Hess } V(X_0)]_{\alpha\beta}$ and M_A^2 is the gauge-field endomorphism, which vanishes in the unbroken H phase. The background-field derivation of (C2) follows Abbott's method [34]; heat-kernel methods are reviewed in [35].

Because Gaussian path integrals satisfy [16, 34]

$$\int D\varphi e^{-\frac{1}{2}\varphi^T \Delta \varphi} = (\det \Delta_\phi)^{-\frac{1}{2}}, \quad \int D\bar{c} Dc e^{\bar{c} \Delta_{\text{ghost}} c} = \det \Delta_{\text{ghost}},$$

the one-loop effective action is

$$\Gamma^{(1)} = \frac{1}{2} \log \det(-\Delta_\phi) - \log \det(-\Delta_{\text{ghost}}) + \dots$$

The overall minus sign for the ghost determinant matches the background-field convention of [34]. Using the proper-time representation [36],

$$\log \det(-\Delta) = - \int_{\mu^{-2}}^{\Lambda^{-2}} \frac{d\tau}{\tau} \text{Tr}[e^{-\tau\Delta}] = - \sum_{n=0}^{\infty} a_{2n}(\Delta) \frac{\Lambda^{2-2n} - \mu^{2-2n}}{2-2n},$$

and the Seeley-DeWitt coefficients of Table C1, the a_2 term from the $N_s = 32$ Goldstone scalars yields

$$\Gamma^{(1)} \supset \frac{1}{2} M_{\text{Pl, scalar}}^2 \int d^4x \sqrt{-g} R, \quad M_{\text{Pl, scalar}}^2 = \frac{N_s f^2}{(4\pi)^2} \log \frac{\Lambda^2}{\mu^2}.$$

□

Table C1 Lowest Seeley-DeWitt coefficients (Euclidean signature) for minimal Laplace-type operators $\Delta = -\nabla^2 + \mathcal{E}$. All curvature tensors refer to $g_{\mu\nu}$.

Field	a_0	a_2	a_4
Real scalar	1	$\frac{1}{6} R$	$\frac{1}{180} (R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - R_{\mu\nu} R^{\mu\nu}) + \frac{1}{72} R^2$
Weyl fermion	2	$-\frac{1}{6} R$	$\frac{7}{720} R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - \frac{1}{180} R_{\mu\nu} R^{\mu\nu} + \frac{1}{144} R^2$
Gauge vector	2	$\frac{1}{6} R$	$\frac{1}{30} R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - \frac{1}{30} R_{\mu\nu} R^{\mu\nu}$
Vector ghost (Grassmann)	2	$\frac{1}{6} R$	$-\frac{11}{180} R_{\mu\nu\rho\sigma}^2 + \frac{1}{30} R_{\mu\nu}^2 - \frac{1}{60} \square R$

Proof of Theorem 26 Expanding the one-loop effective action (8) about $X = X_0 + \varphi$, $\varphi \in \mathfrak{p} = T_{X_0}\mathcal{M}$, and retaining only $\mathcal{O}(\varphi^2)$ terms yields (9):

$$\Gamma^{(1)}[X_0 + \varphi] = \Gamma^{(1)}[X_0] + \frac{1}{2} \int d^4x \varphi^I [\text{Hess } V(X_0)]_{IJ} \varphi^J + \dots,$$

where $I, J \in \{1, \dots, 32\}$ label a basis of $\mathfrak{p}$.

The tree potential V of Theorem 11 is E_6 -invariant and vanishes on the whole orbit $\mathcal{M}$; hence $\text{Hess } V(X_0) = 0$ identically at tree level, and all 32 coset scalars are exact Goldstone bosons. The gauge and Yukawa couplings of (6) respect only $H = \text{Spin}(10) \times \text{U}(1)$, so the one-loop effective potential V_{eff} is H -invariant but not E_6 -invariant; its Hessian therefore commutes with the H -action. Since $\mathfrak{m} \cong \mathfrak{p}$ carries an irreducible representation of H of complex type (Lemma 21), an invariant endomorphism lies in the commutant $\mathbb{C} = \text{span}\{\mathbf{1}, J\}$; symmetry of the Hessian with respect to the invariant metric excludes the antisymmetric J -part, so it is a multiple of the identity,

$$\text{Hess } V_{\text{eff}}(X_0) = m^2 \mathbf{1}_{32}.$$

Assuming the one-loop correction is stabilizing, $m^2 = f^2 \mu^2 > 0$ is the single common eigenvalue fixed by the FRG flow; no H -invariant $4 + 28$ splitting of the mass matrix exists. On a non-degenerate spacetime-filling background the four directions $\mathfrak{p}_0 = \Im d\phi$ of Lemma 16 are singled out by the frame condensate $\langle \partial_\mu \phi^a \rangle = e_\mu^a$; their gradients constitute the tetrad, which is promoted to the gravitational field by the induced Einstein-Hilbert term (Theorem 25) and does not propagate as an independent scalar. The remaining 28 transverse fluctuations $\varphi^{\hat{\alpha}} \in \mathfrak{p}_m$ obey $(\square + m^2) \varphi^{\hat{\alpha}} = 0$ with $m^2 = f^2 \mu^2$. $\square$

Appendix D Functional Renormalization-Group Details

This appendix records the functional renormalization-group machinery behind § 5: the exact flow equation and the regulator, the truncation and projection rules, and the derivations of the one- and two-loop β -functions quoted in the main text.

We adopt the background-field functional RG [37, 38]. For the coarse-grained effective action Γ_k the exact flow equation reads

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr} \left[\left(\Gamma_k^{(2)} + R_k \right)^{-1} \partial_k R_k \right],$$

where $\Gamma_k^{(2)}$ is the Hessian with respect to the collective fields $\Phi = (\phi^\alpha, A_\mu^i, \psi)$ and R_k is an IR regulator that suppresses modes with $p^2 \lesssim k^2$. We work in background Lorenz gauge for the $\text{Spin}(10) \times \text{U}(1)$ gauge sector, add the corresponding Faddeev-Popov determinant, and impose standard BRST sources. The background-field identities then guarantee that Γ_k remains gauge invariant at every k .

All threshold functions entering the flow are evaluated with

$$R_k(p^2) = Z_k(k^2 - p^2) \theta(k^2 - p^2),$$

where Z_k is the wave-function renormalization of the field in question. Litim's choice maximizes the gap in the inverse propagator and yields analytic expressions for the threshold integrals while preserving locality of Γ_k [39].

Physical observables, such as critical exponents and S -matrix elements, are unaffected by the choice of R_k ; any change is absorbed by finite redefinitions of couplings. Explicit regulator-variation checks for the present truncation reproduce the standard $\sim 1\%$ scheme uncertainty.

The truncation employed in § 5 keeps only the lowest-dimensional operators that mix under the RG flow,

$$\Gamma_k = \int d^4x \left[\frac{1}{2f_k^2} \partial_\mu \phi^\alpha \partial^\mu \phi_\alpha + \frac{1}{4g_k^2} F_{\mu\nu}^i F^{i\mu\nu} + \frac{y_k}{2} \phi^\alpha \bar{\psi} \psi \right] + \Gamma_k^{\text{grav}}[\eta_{\mu\nu}],$$

where ϕ^α denotes the 32 coset scalars, $F_{\mu\nu}^i$ the $\text{Spin}(10) \times \text{U}(1)$ field strength and ψ the chiral matter multiplet of Table 1. Higher-derivative and multi-scalar operators enter only at subleading order in the derivative expansion and do not affect the universal data extracted here. We write $t = \log k$ and denote

$$\eta_f \equiv -\partial_t \log f_k^2, \quad \eta_g \equiv -\partial_t \log g_k^2, \quad \eta_y \equiv -\partial_t \log y_k^2, \quad \eta_M \equiv -\partial_t \log M_{\text{Pl}}^2(k).$$

Following [40], we absorb canonical dimensions by

$$\hat{g}^2 = k^2 \frac{g_k^2}{f_k^2}, \quad \hat{y}^2 = y_k^2, \quad \xi = \frac{k^2}{f_k^2}, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2(k)}. \quad (\text{D3})$$

Differentiation of (D3) gives the projection rules

$$\begin{aligned} \beta_{\hat{g}^2} &= (2 - \eta_f + \eta_g) \hat{g}^2, & \beta_{\hat{y}^2} &= (-\eta_y) \hat{y}^2, \\ \beta_\xi &= (2 - \eta_f) \xi, & \beta_\kappa &= (2 + \eta_M) \kappa. \end{aligned} \quad (\text{D4})$$

Hence the task reduces to computing the four anomalous dimensions $\{\eta_f, \eta_g, \eta_y, \eta_M\}$ from the trace on the right of the Wetterich equation.

For a simple algebra $\mathfrak{g}$ with adjoint index C_A , fundamental index T_R , n_F Weyl fermions and n_S real scalars the background-field flow reproduces the standard minimal-subtraction result

$$\beta_{g^{-2}} = \frac{b_0}{48\pi^2}, \quad b_0 = \frac{11}{3}C_A - \frac{4}{3}T_R n_F - \frac{1}{6}T_R n_S. \quad (\text{D5})$$

For the E_6 matter content $(C_A, T_R, n_F, n_S) = (12, 1, 16, 32)$, we obtain $b_0 = 52/3$. The 32 coset scalars are (pseudo-)Goldstone modes of the breaking $E_6 \rightarrow \text{Spin}(10) \times \text{U}(1)$; at one loop their two-point function is protected by the coset symmetry, hence

$$\eta_f = -\partial_t \log f_k^2 = 0. \quad (\text{D6})$$

28 of the 32 scalars acquire a common mass $m^2 = f^2 \mu^2$ (Theorem 26). Their one-loop contribution is the functional determinant $(1/2) \text{Tr} \log [-D^2 + m^2]$. Using the

heat-kernel expansion

$$\text{Tr } e^{-s(-D^2)} = (4\pi s)^{-2} \sum_{n \geq 0} s^n a_{2n}(\Delta),$$

and the Seeley-DeWitt entries in [Table C1](#), the term proportional to a_2 induces

$$\partial_t \Gamma_k \supset \frac{5}{48\pi^2} f^2 R \sqrt{-\eta} \implies \beta_{M_{F1}^2} = -\frac{5}{48\pi^2} f^2. \quad (\text{D7})$$

The minus sign follows from $\mu \partial_\mu \log(\Lambda^2/\mu^2) = -2$ after identifying $k = \mu$. The Yukawa screening coefficient entering η_y is $b_y = 2C_F/(16\pi^2)$, $C_F = 9/4$, so that

$$\eta_y = 2b_y \hat{y}^2. \quad (\text{D8})$$

Collecting [\(D5\)](#)–[\(D8\)](#), one finds

$$\eta_f = 0, \quad \eta_g = \frac{b_0}{24\pi^2} \hat{g}^2, \quad \eta_y = 2b_y \hat{y}^2, \quad \eta_M = +\frac{5}{48\pi^2} \kappa.$$

Inserting these values into the projection identities [\(D4\)](#) reproduces the one-loop β -functions [\(11\)](#) quoted in the main text.

Coset-scalar self-interactions and Yukawa diagrams enter at two loops. A background-field expansion with the Litim regulator produces the coefficients listed in [Table D2](#), which gives both their group-invariant formulas and their numerical values for the E_6 field content.

Table D2 Two-loop β -function coefficients for the E_6 field content ($C_A = 12$, $C_F = 9/4$, $T_R = 1$, $n_F = 16$, $n_S = 32$).

Coefficient	Symbolic expression	E_6 value
A	$\frac{T_R}{192\pi^4}$	5.35×10^{-5}
b_0	$\frac{\frac{11}{3}C_A - \frac{4}{3}T_R n_F - \frac{1}{6}T_R n_S}{16\pi^2}$	1.10×10^{-1}
b_1	$\frac{34C_A^2 - 4C_A T_R n_F - \frac{5}{3}C_A T_R n_S - 4C_F T_R n_F}{(16\pi^2)^2}$	1.34×10^{-1}
c_1	$-\frac{3C_F}{16\pi^2}$	-4.27×10^{-2}
c_2	$\frac{2C_F - \frac{3}{2}C_A}{16\pi^2}$	-8.55×10^{-2}
b_y	$\frac{2C_F}{(16\pi^2)}$	2.85×10^{-2}

Let $(\xi_*, \hat{g}_*^2, \hat{y}_*^2, \kappa_*)$ denote a simultaneous zero of the algebraic β -system, without assuming that the zero lies in the positive-coupling domain. Linearizing $\partial_t \delta \mathbf{g} = M \delta \mathbf{g}$, one obtains

$$M = \begin{pmatrix} -2 & -A \xi_* & -b_0 \xi_* & 0 \\ 0 & 2\hat{g}_*^2 b_1 & 2\hat{g}_*^2 c_1 & 0 \\ 0 & c_2 \hat{y}_*^2 & 2b_y \hat{y}_*^2 & 0 \\ 0 & 0 & 0 & -2 - \frac{5}{24\pi^2} \kappa_* \end{pmatrix}. \quad (\text{D9})$$

For the coefficients of Table D2, the nontrivial gauge–Yukawa branch has $\hat{g}_*^2 < 0$ and $\hat{y}_*^2 < 0$. It is therefore excluded from the physical positive-coupling regime. The stability matrix is retained here only as an audit of the printed truncation; it is not evidence for an autonomous positive-coupling asymptotic-safety fixed point.

Within the effective coset regime below $\epsilon < 1/6$, the one-loop $2 \rightarrow 2$ amplitude for canonically normalized coset scalars has the form

$$\mathcal{A}^{(1)} = \frac{\hat{g}^4}{16\pi^2} \log \frac{s}{t} + \text{cyclic}(s, t, u),$$

which is finite inside the effective truncation. See [40] for method details.

Appendix E Diagonalization of the 27×27 Scalar Stability Block

The classically marginal quartic couplings of the coset scalars can be organized as a vector $(\lambda^a)_{a=0}^{26}$ transforming in the fundamental $\mathbf{27}$ of E_6 . Linearizing their effective flow within the $\epsilon < 1/6$ coset regime gives

$$\partial_t \lambda^a = M_b^a \lambda^b, \quad M_b^a = \frac{\partial \beta^a}{\partial \lambda^b}.$$

Using group-theoretic projectors and the identities of [4], the matrix M is block-diagonal and, for the E_6 field content, diagonal:

$$M = \text{diag}(-2.00, -1.89, -1.57, -1.18, -0.92, -0.77, -0.44 (\times 21)),$$

all eigenvalues being negative. Hence no additional relevant directions arise inside this effective scalar block before the heterotic handoff.

Declarations

The author declares no competing interests. Code for reproducing the results and associated datasets are available at <https://github.com/belna-inc/ucft>. No external funding is reported.

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